

Web Development Using HTML

Internet Programming I: Chapter 2



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY
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The core Web development technologies



- HTML, CSS and JavaScript are the basic building block and core web development technologies.



Structure

- Create the structure of the website



Presentation

- Stylize the layout of the website



Behavior

- Add interactivity to the website



What is the Difference



source: <https://admin.brytdesigns.com/>



HTML
Hypertext Markup Language

Create the structure

- Controls the layout of the content
- Provides structure for the web page design
- The fundamental building block of any web page



CSS
Cascading Style Sheet

Stylize the website

- Applies style to the web page elements
- Targets various screen sizes to make web pages responsive
- Primarily handles the "look and feel" of a web page



Javascript

Increase interactivity

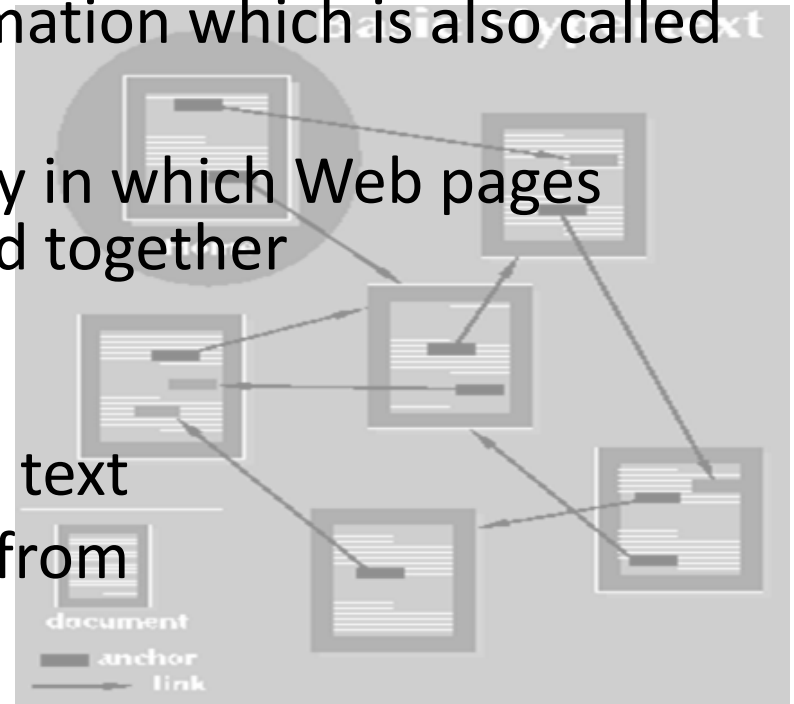
- Adds interactivity to a web page
- Handles complex functions and features
- Programmatic code which enhances functionality

Hypertext

- HTML Stands for Hypertext Markup Language
- Hypertext basically refers to Non-sequentially linked pieces of text or other information which is also called “Nodes”
- More specifically it is the way in which Web pages (HTML documents) are linked together

Goal

Allow to access and read text and other information from multiple perspectives



Markup Language



- Markup originally refers to the handwritten indicators on an author's manuscripts.
- The idea of markup Notify a typesetter the layout of a document and the typeface to use
- A markup language is a set of characters or symbols that define a document's logical structure.
- In other words, it refers to the use of set of characters within a piece of information that can be used to process or identify that information in a particular way
- It composed of set of symbols called markup tags.

HTML (Hypertext Markup Language)



- It is a markup language which based on Standard Generalized Markup Language (SGML)
- SGML is a standard for specifying a markup language or tag set
- SGML Itself is not a document language, but a description of how to specify one and create a document type definition (DTD)
- It define the structure of information on the Web page
- It doesn't describe the actual presentation of a document
- It tells the web browser what content to display.
- Use a pre-defined set of tags to identify the webpage content types .
- It is not a programming language.

HTML

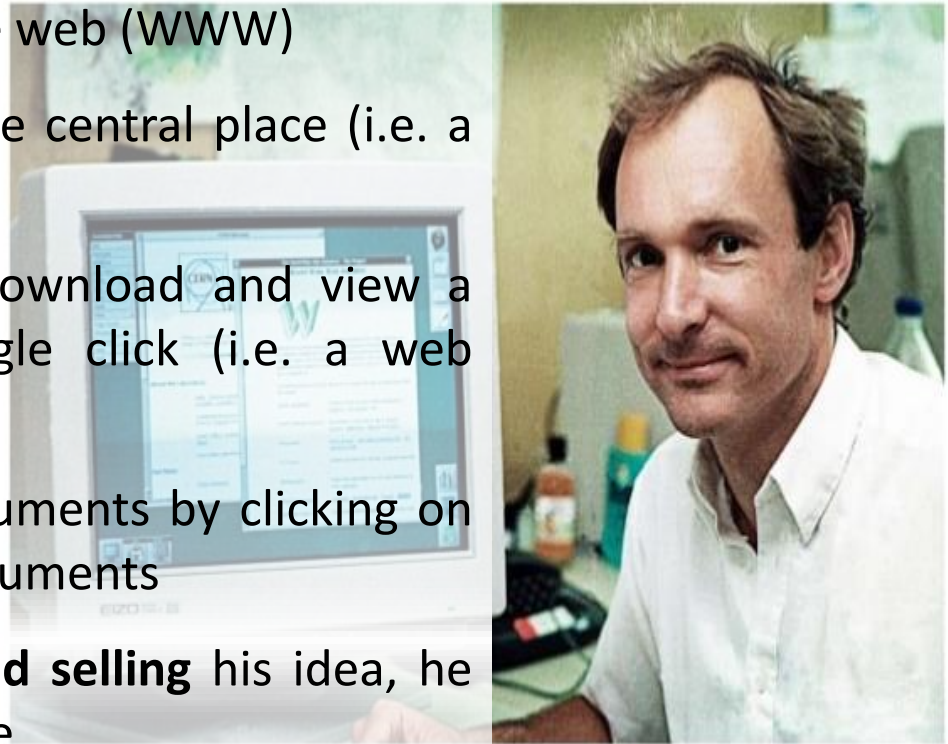


- HTML is a language made up of elements, which can be applied to pieces of text to give them different meaning in a document
 - Is it a paragraph?
 - Is it a bulleted list?
 - Is it part of a table?
- Structure a document into logical sections
 - Does it have a header?
 - Three columns of content?
 - A navigation menu?
- Embed content such as images and videos into a page

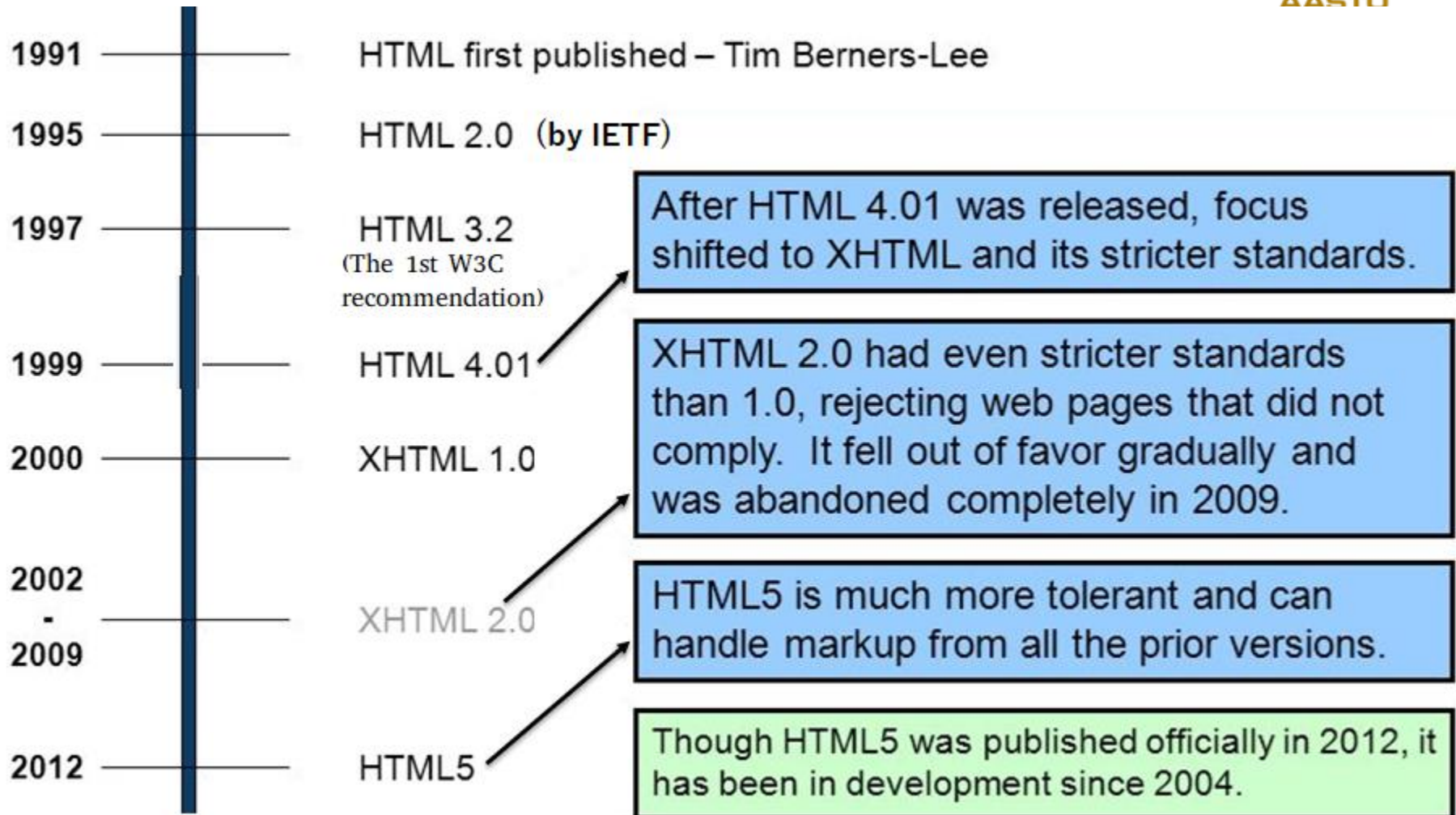
The evolution of HTML



- Before 1990 there was no easy way to find, download and view documents over the internet
- HTML, initially invented by in 1991 by Tim Berners Lee
- **Tim** invented a system – the web (WWW)
 - Store documents in one central place (i.e. a web server)
 - Make it possible to download and view a document with a single click (i.e. a web browser)
 - Allow to find new documents by clicking on “links” inside other documents
 - Instead of **licensing and selling** his idea, he made free for every one



The evolution of HTML cont'd

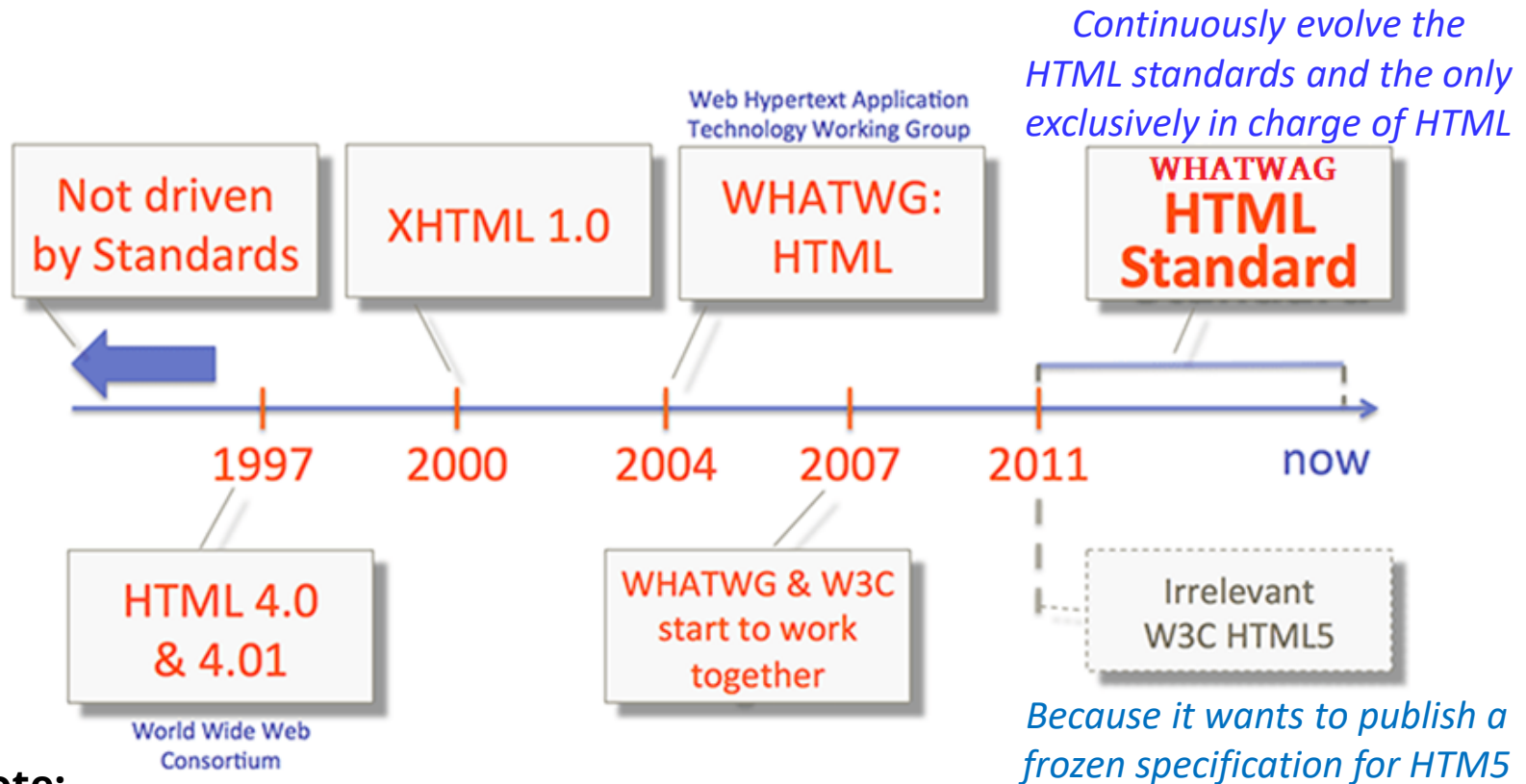


HTML Standardization history



- Before 1997, there were no HTML standards, so browsers basically did whatever they wanted
- Particularly 1993 – 1997 (The “War” Years), HTML was being defined by the elements that browser software developers chose to implement
- They invented new tags or implemented the same tags differently which was kind of the wild west of the web
- Around 1997, the W3C came up with the first standard that browsers started to pay attention to
- The W3C defined HTML version 4 (or HTML4) which it shortly thereafter updated to HTML4.01

HTML Standardization history cont'd



Note:

- **W3C** is the main international standards organization for the Web
- **WHATWG** – established in 2004 by the browser vendors

HTML5



- Current version of standard HTML which was defined in 2012 by WHATWAG
- It is cross-platform. It will display content on a desktop computer, laptop, a tablet, smartphone, a notebook or a Smart TV
- All major browsers support most of the HTML5 elements
- Key features
- Multimedia elements
- Graphics elements
- Offline support
- Semantics markups

Basic Structure of HTML Document



`<!DOCTYPE>`

- *Tell the browser the type of HTML standard*
- *Must appear first*

`<html>`

`<head>`

`</head>`

`<body>`

`</body>`

`</html>`

- *Contain html document meta info, links, style sheet and scripts etc.*
- *These are non web displayable info*

- *Contains the content that displays in the browser*

Basic HTML Page Example



```
<html>
```

```
<head>
```

```
<title>Hello World </title>
```

```
</head>
```

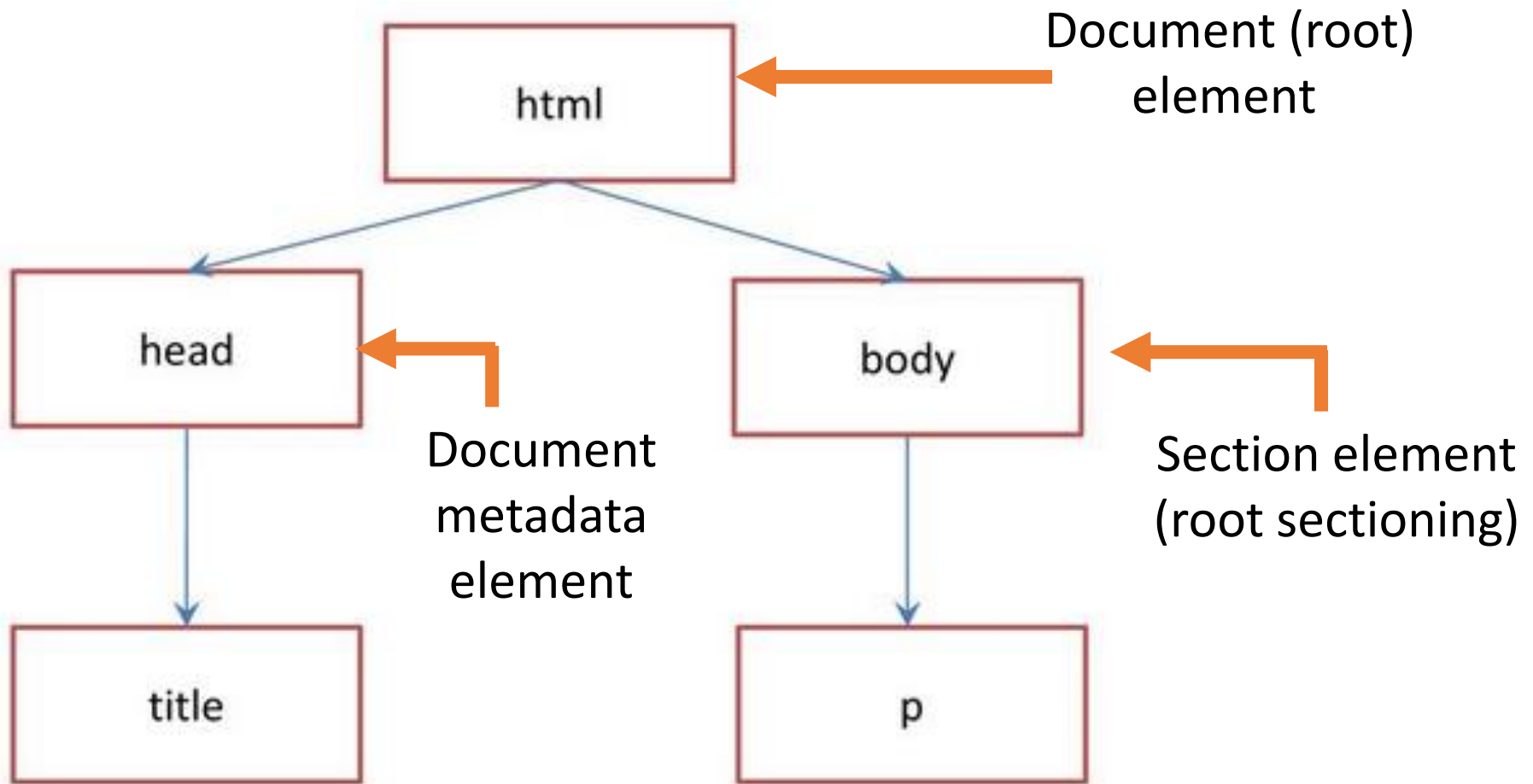
```
<body>
```

```
<p>Welcome to the World </p>
```

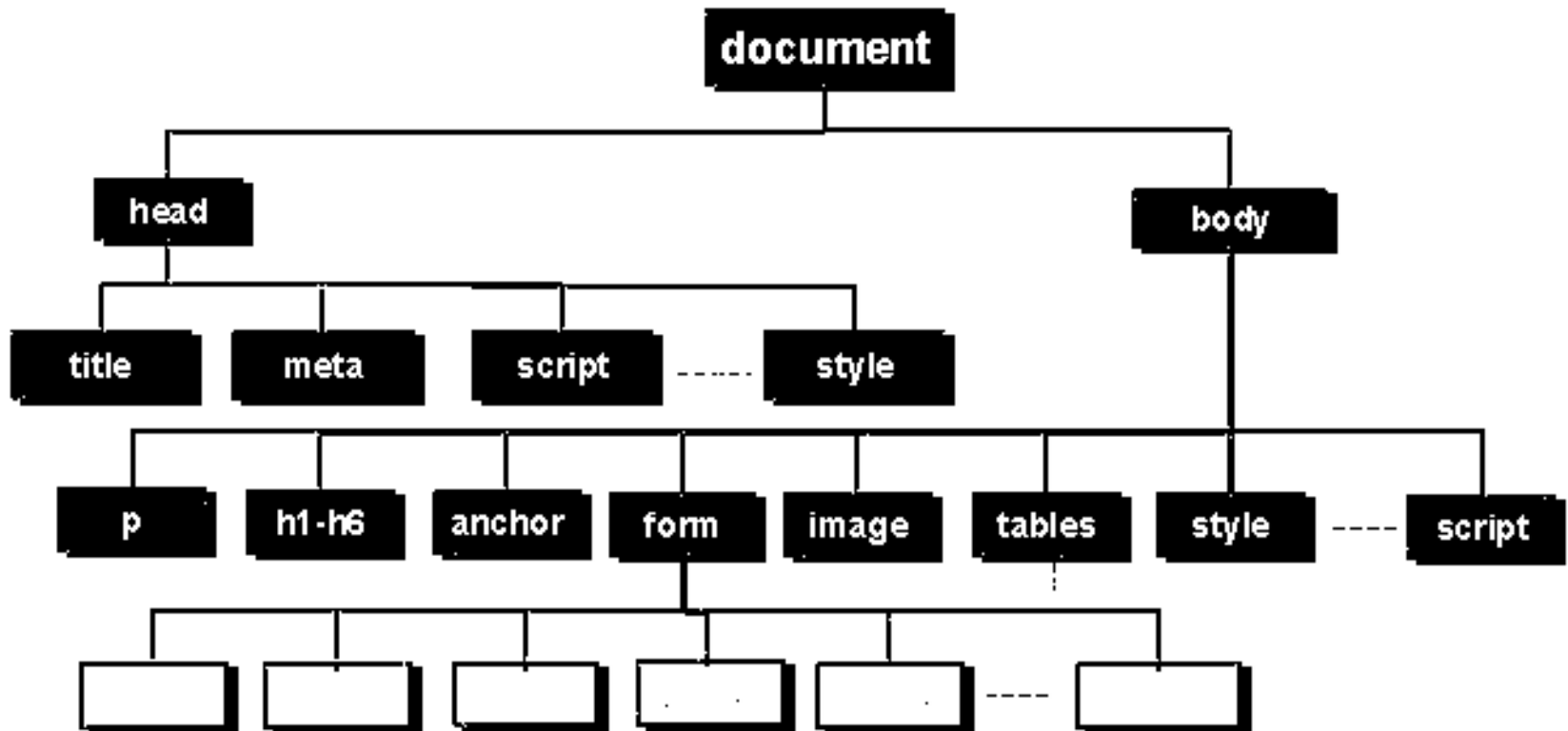
```
</body>
```

```
</html>
```

Tree structure of HTML Document



Tree structure of HTML Document cont'd



Tree structure of HTML Document



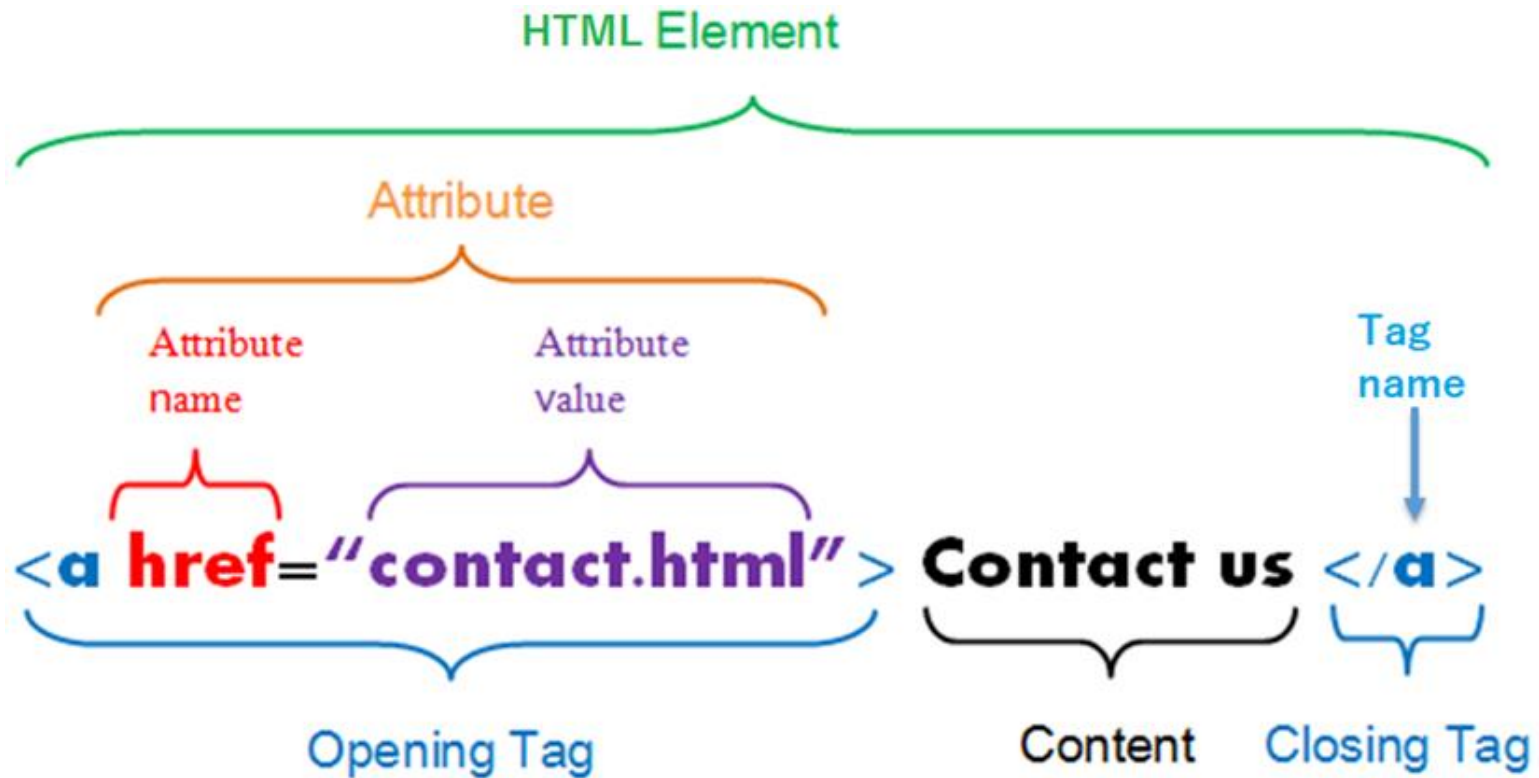
Inside the body section

- Text formatting
- Links
- Table
- Form
- Images and multimedia
- Embedded elements
- Interactive elements
- Graphics etc.

Inside the head section

- Title
- Style
- Script /no script
- Meta
- Link
- base

Anatomy of HTML Elements/Tags



- **Syntax:**

`<tag attribute = "Value"> Content </tag>`

Anatomy of HTML Elements/Tags cont'd



HTML elements

- Represent some kind of structure
- It is a combination of a tag and its character data (content)
- HTML can be Empty element or Nested element also

Nested Element

- An element that contain other HTML elements

Empty Element

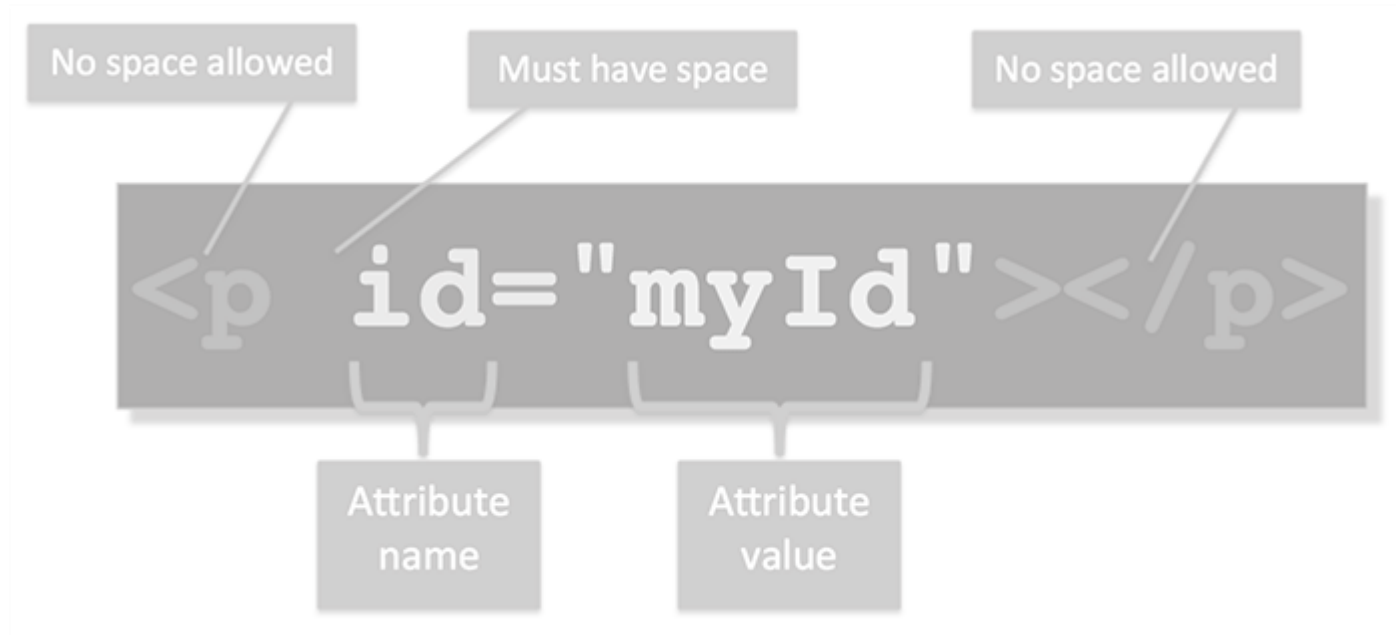
- An element with no character data (content)
- Also called Non-container tags or self closing tags
- Syntax: `<self-closing-tag-name/>`

Anatomy of HTML Elements/Tags cont'd



- HTML Tags
 - Used to markup the text (content)
 - Most of the tags come in as start and end tag pairs
 - However, empty element tags doesn't come in pairs
 - The start tag may contain an attribute
- HTML attributes
 - Defines a property for an elements
 - It is a way to describe the tags
 - It must enclosed within the start tag
 - It comes in a name/value pairs
 - The value should enclosed within double quote
 - An element can have one or more attributes
 - Global attributes - common to all HTML elements

Simple Rule



Also:

- Don't forget the End tag.
- Even though HTML is case insensitive use lowercase Tags

HTML elements reference



- The HTML elements are grouped by function to help you find what you have in mind easily
- For details of each element under the below mentioned group just **ctrl +click** on the element category

- [Document metadata](#)
- [Sectioning root](#)
- [Content sectioning](#)
- [Text content](#)
- [Inline text semantics](#)
- [Image and multimedia](#)
- [Embedded content](#)
- [SVG and MathML](#)
- [Scripting](#)
- [Demarcating edits](#)
- [Table content](#)
- [Forms](#)
- [Interactive elements](#)
- [deprecated elements](#)

Content Model in HTML



- Content Model refers to the **set of rules** that define what type of content each element is allowed to have.
- Mostly, this translates into what other elements are allowed to be nested inside which other elements.
- Prior to the modern HTML specification, HTML elements were either **block-level or inline elements**
- Modern HTML specification split these two content models into seven basic models
- However, in practical terms, you can still think of those seven models as falling into the same traditional categories: block-level and inline elements

Content Model in HTML cont'd



Block-level Elements

- By default, Block-level elements render (displayed) to begin on a new line (i.e. the browser will automatically place the contents of that element on a new line in the flow of the document)

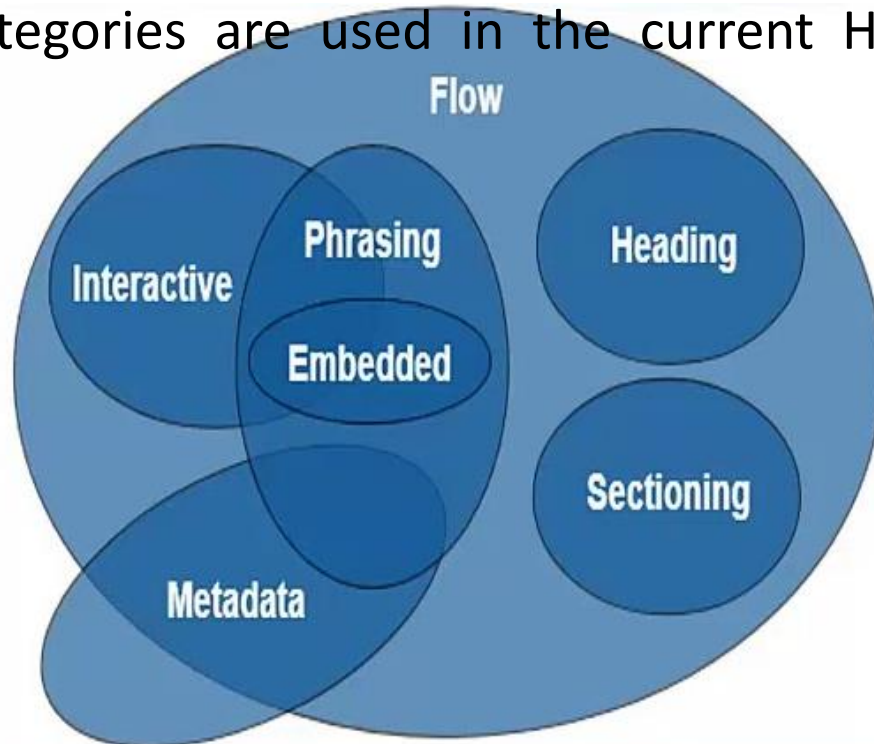
Inline Elements

- Render on the same line by default
- Having new line characters in the content or between the tags in your code won't make any difference
- All those space characters get translated into a single space anyway and the whole bunch of inline elements still be displayed on the same line
- Inline elements are restricted to only contain other inline elements

Modern HTML Content Models

- Each elements in HTML5 falls into zero or more categories that group elements with similar characteristics
- The following broad categories are used in the current HTML specification:

- Metadata content
- Flow content
- Sectioning content
- Heading content
- Phrasing content
- Embedded content
- Interactive content



Document structure elements



This meta element defines a description of your page:

```
<META name="description" content="Free Web tutorials on HTML, CSS, XML, and XHTML" />
```

This meta element defines keywords for your page: (for search engine)

```
<META name="keywords" content="HTML, DHTML, CSS, XML, XHTML, JavaScript" />
```

This demonstrates how to redirect a user if your site address has changed:

```
<META http-equiv="Refresh" content="5;url=http://www.w3schools.com" />
```

This meta element defines character set:

```
<META http-equiv="Content-Type" content="text/html; charset=UTF-8" />
```

Text Markup Elements



- Heading elements
 - In the same way that a book has the main title, chapter titles, and subtitles, an HTML document can too.

`<h1>How to Make a Cake</h1>` ← - - - - - Main heading

`<h2>Ingredients</h2>` ← - - - - - Sub heading

`<h3>Ingredients for Sponge</h3>` } Sub sub heading

`<h3>Ingredients for Toppings</h3>` }

`<h2>Instructions</h2>`

`<h3>Preparation</h3>`

`<h3>Process</h3>`

`<h2>Serving the Cake</h2>`

Text Markup Elements cont'd



- Heading elements allow you to specify that certain parts of your content are headings
- HTML contains 6 section heading levels, <h1> to <h6>
- The heading element importance level
 - <h1> define the most important (highest) heading level
 - <h6> define the lowest (less important) headings
- Commonly 3 to 4 heading elements are used at most
- The heading Tags are block level elements.
- It's content model categories are flow content and heading content
- These elements only include the global attributes

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

Text Markup Elements cont'd



- Heading information
 - Allow search engines to index the structure and content of your web pages
 - Users often skim a page by its headings and determine the content layout of web page automatically.
 - Also it is a common navigation technique for users of screen reading software to jump from heading to heading and quickly determine the content of the page.
- As a result,
 - Avoid skipping heading levels: always start from <h1>, followed by <h2> and so on
 - Use only one <h1> per page or view (considered a best practice)
 - <h1> should concisely describe the overall purpose of the content.
 - It is important to use headings to show the document structure.
 - Avoid using heading elements to resize text (i.e. to make text BIG or bold).

Text Markup Elements cont'd



Paragraph elements

- HTML paragraphs any structural grouping of related content, such as texts, images or form fields
- The <p> HTML element represents a paragraph
- It belongs flow content category
- Paragraphs are block-level elements and
 - Always starts on a new line
 - Browsers automatically add white space (margin) before and after a paragraph
- This element only includes the global attributes
- Breaking up content into paragraphs helps make a page more accessible
- Allow screen-readers to provide shortcuts that let their users skip to the next or previous paragraph, and skim the content

Text Markup Elements cont'd



- **HTML <p> content display**

- Large or small screens, and resized windows will create different results.
- With HTML, you cannot change the display by adding extra spaces or extra lines in HTML code.
- The browser will automatically remove any extra spaces and lines.

- **Note:**

- Don't use empty <p> which is problematic for people who use screen-reading technology.

```
<p>
This paragraph contains a lot of
in the source code, but the browser
ignores it.
</p>

<p>
This paragraph
contains      a lot of spaces
in the source   code,
but the   browser
ignores it.
</p>
```

Output of the example

This paragraph contains a lot of lines in the source code, but the browser ignores it.

This paragraph contains a lot of spaces in the source code, but the browser ignores it.

Text Markup Elements cont'd



<pre> elements

- Represents preformatted text which is to be presented exactly as written in the HTML file
 - The text is typically rendered using a non-proportional, or "monospaced", font
 - Unlike <p> element, it preserves whitespace both spaces and line breaks:
- Provide an alternate description for any images or diagrams
- It is a block-level element which belongs to flow content category

<pre>

< I'm an expert in my field. >

\ ^ ^
 \ (oo) \ _____
 (_) \) \ \
 || -----w ||
 || ||

</pre>

Output

< I'm an expert in my field. >

\ ^ ^
 \ (oo) \ _____
 (_) \) \ \
 || -----w ||
 || ||

Text Markup Elements cont'd



Line break elements (<hr> and
)

<hr/>

- Represents a thematic break between paragraph-level elements:
- E.g., a change of scene in a story, or a shift of topic within a section.
- Displayed as a horizontal rule in visual browsers

- Produces a line break in text (carriage-return)
- It is useful for writing a poem or an address, where the division of lines is significant
- Only enter line breaks, not to add space between paragraphs

- Both <hr> and
 support both global attributes and event attributes
- Both are block-level elements,
- <hr> is flow content whereas
 is phrasing content

Text Markup Elements cont'd



```
<p> §1: The first rule of Fight Club is: You  
do not talk about Fight Club. </p>  
<hr/>  
<p> §2: The second rule of Fight Club is:  
Always bring cupcakes. </p>
```

```
<h3>A Poem</h3>
```

```
<p> O'er all the hilltops<br>  
Is quiet now,<br>  
In all the treetops<br>  
Hearest thou<br>  
Hardly a breath;<br>  
The birds are asleep in the trees:<br>  
Wait, soon like these<br>  
Thou too shalt rest.  
</p>
```

§1: The first rule of Fight Club is: You do
not talk about Fight Club.

Output

§2: The second rule of Fight Club is:
Always bring cupcakes.

A Poem

Output

O'er all the hilltops
Is quiet now,
In all the treetops
Hearest thou
Hardly a breath;
The birds are asleep in the trees:
Wait, soon like these
Thou too shalt rest.

Text Markup Elements cont'd



List elements

- HTML offers web developers mechanisms to group a set of related items in lists
- HTML lists are used to present list of information in well-formed and semantic way
- There are three main types of list in HTML, each one has a specific purpose

➤ *Used to create a list of related items, in no particular order*

Unordered List

- The first item
- The second item
- The third item
- The fourth item

➤ *Used to create a list of terms and their descriptions.*

Ordered List

1. The first item
2. The second item
3. The third item
4. The fourth item

➤ *Used to create a list of related items, in a specific order.*

Description List

The first item
Description of item
The second item
Description of item

Text Markup Elements cont'd



No	List element	Attribute(s)	Attribute value(s)
1		type	Circle / disc / square / triangle
2		reversed	Boolean
		type	1 (default), A / a / I / i
		start	integer - indicates the current ordinal value of list item
3		value	Integer - start counting from for the list items.
		type	1 (default) A / a / I / i
4	<dd>	nowrap	Yes/no (default), If the value is set to yes, the definition text will not wrap

Text Markup Elements cont'd



Example: Ordered list

```
<h2>Hot Drink</h2>
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
  <li>macchiato</li>
</ol>
```

```
<h2>Hot Drink</h2>
<ol type="a" start = "4">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
  <li>macchiato</li>
</ol>
```

```
<h2>Hot Drink</h2>
<ol type="a" reversed = "1">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
  <li>macchiato</li>
</ol>
```

Hot Drink

1. Coffee
2. Tea
3. Milk
4. macchiato

Hot Drink

- d. Coffee
- e. Tea
- f. Milk
- g. macchiato

Hot Drink

- d. Coffee
- c. Tea
- b. Milk
- a. macchiato

Text Markup Elements cont'd

Example: Unordered list

```
<ul>
  <li>Fruit
    <ul>
      <li>Bananas</li>
      <li>Apples
        <ul>
          <li>Green</li>
          <li>Red</li>
        </ul>
      <li>Pears</li>
    </ul>
  <li>Vegetables</li>
  <li>Meat</li>
</ul>
```

- Fruit
 - Bananas
 - Apples
 - Green
 - Red
 - Pears
- Vegetables
- Meat

Nested list

```
<ol>
  <li>Fruit
    <ul type = "disc">
      <li>Bananas</li>
      <li>Apples
        <ul type = "circle">
          <li>Green</li>
          <li>Red</li>
        </ul>
      <li>Pears</li>
    </ul>
  <li>Vegetables</li>
  <li>Meat</li>
</ol>
```

1. Fruit
 - Bananas
 - Apples
 - Green
 - Red
 - Pears
2. Vegetables
3. Meat

Text Markup Elements cont'd



Example: Description list

```
<h2>Acronym List</h2>

<dl>
  <dt>HTML</dt>
  <dd>- Hypertext Markup Language</dd>
  <dt>CSS</dt>
  <dd>- Cascading Style Sheet</dd>
  <dt>PHP</dt>
  <dd>- Hypertext Preprocessor</dd>
</dl>
```

Acronym List

HTML

- Hypertext Markup Language

CSS

- Cascading Style Sheet

PHP

- Hypertext Preprocessor

Text Markup Elements cont'd



Text formatting elements

- HTML provides several tags that you can use to make some text on your web pages to appear differently than normal text
- Formatting elements were designed to display special types of text:

Tag	Description
<u></u>	Defines bold text
<u><big></u>	Defines big text
<u></u>	Defines emphasized text
<u><i></u>	Defines italic text
<u><small></u>	Defines small text
<u></u>	Defines strong text

Tag	Description
<u><sub></u>	Defines subscripted text
<u><sup></u>	Defines superscripted text
<u><s></u>	Renders text with a strikethrough but not important when indicating document edits
<u><u></u>	Deprecated. But in use for different purpose

Text Markup Elements cont'd



Demarcating and Quotation Elements

Tag	Description
<ins>	Defines inserted text
	Defines deleted text
<mark>	Marked text
<samp>	sample output of a computer program
<code>	To show the computer code - Like a programming code
<kbd>	keyboard input text
<var>	To display the text as a variable - Like Programming variable

Text Markup Elements cont'd



Example

`<i>Italic</i>`

Italic

`<b>Bold</b>`

Bold

`<em>Emphasized</em>`

Emphasized

`<strong>Strong</strong>` **Strong**

`<small>small</small>`

small

`<del>Deleted</del>`

~~Deleted~~

`<ins>Inserted</ins>`

Inserted

`v<sub>f</sub>`

v_f

`a<sup>2</sup>`

a^2

`<mark>Marked</mark>`

Marked

w

Text Markup Elements cont'd



HTML Citations and Definition Elements

Tag	Description
<u><abbr></u>	Defines an abbreviation
<u><acronym></u>	Defines an acronym
<u><address></u>	Defines contact information for the author/owner of a document
<u><bdo></u>	Defines the text direction
<u><blockquote></u>	Defines a long quotation
<u><q></u>	Defines a short quotation
<u><cite></u>	Defines a citation
<u><dfn></u>	Defines a definition term

HTML Entity and Comment



- A piece of text ("string") that begins with an ampersand (&) and ends with a semicolon (;)
- Frequently used to display
- Reserved characters (which would otherwise be interpreted as HTML code)
- Invisible characters (like non-breaking spaces).
- Characters that are difficult to type with a standard keyboard.
- Syntax:
 - `&entity_name;` OR `&#entity_number;`
- An entity name is easy to remember but browsers may not support all entity names. Therefore, the support for entity numbers is good.

HTML Entity and Comment cont'd



- HTML reserved character Entities

Character	Description	Entity Name	Entity Number
	non-breaking space	 	
<	less than	<	<
>	greater than	>	>
&	ampersand	&	&
"	double quotation mark	"	"
'	single quotation mark (apostrophe)	'	'
«	Left-pointing double angle quotation mark	«	«
»	Right-pointing double angle quotation mark	»	»

HTML Entity and Comment cont'd



- HTML reserved character Entities

Character	Entity Name	Entity Number	Description
¢	¢	¢	Cent
£	£	£	Pound
¤	¤	¤	General currency
€	€	€	Euro
¢	¢	¢	Cent
©	©	©	Copyright
®	®	®	Registered
™	™	™	Trademark

HTML Comment



- The syntax to add comments to your HTML source is as follow

`<!-- Write your comments
 goes here -->`

- Notice that there is an exclamation point (!) in the start tag, but not in the end tag.
- Used to hide content - comments are not displayed by the browser, but they can help document your HTML source code.
- With comments you can place notifications and reminders in your HTML code and improve code readability.

HTML Comment cont'd



- Example: Comment

```
<!DOCTYPE html>
<html>
<body>

<p>This is a paragraph.</p>
<!--
<p>Look at this cool image:</p>

-->
<p>This is a paragraph too.</p>

</body>
</html>
```

This is a paragraph.

This is a paragraph too.

Web Development Using HTML

Internet Programming I: Chapter 2 – Part II



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Link elements

- A hyperlink (or link) is a word, group of words, or image that we can click on to jump to a new document or a new section within the current document
- Links are specified in HTML using the anchor (<a>) tag
- The <a> tag can be used in two ways:
 - To create a link to another document, by using the **href** attribute
 - To create a bookmark inside a document, by using the **name** attribute
- **Syntax :** Link text

where “*href*” attribute can point to:

- Outside document
- Another document within the same site
- Another part of document within the same file

Link elements cont'd



Link to external document

- Example:

- `<a href="http://www.w3schools.com/">Visit W3Schools</a>`

- Attribute “target” specifies where to open the document
- The target attribute can have one of the following values:
 - `_self` - Default. Opens the document in the same window/tab as it was clicked
 - `_blank` - Opens the document in a new window or tab
 - `_parent` - Opens the document in the parent frame
 - `_top` - Opens the document in the full body of the window
- Example below opens linked document in a new browser window
 - `<a href="http://www.w3schools.com/" target="_blank">W3Schools</a>`
- An example of a link to document within the same site
 - `<a href="/HR/admin_peple.html">Administrative Staff</a>`

Image Element

- Images enhance visual appearance of the web pages by making them more interesting and colorful
- The `<img>` tag is used to embed images in the HTML documents
- It is an empty element and contains attributes only

Syntax:

(1) **img** is an image tags which embed a given /picture/image

(2) **src** attribute is short for **source** and tells the tag which image to load by specifying it's path

(3) **alt** attribute is short hand for *alternative text*

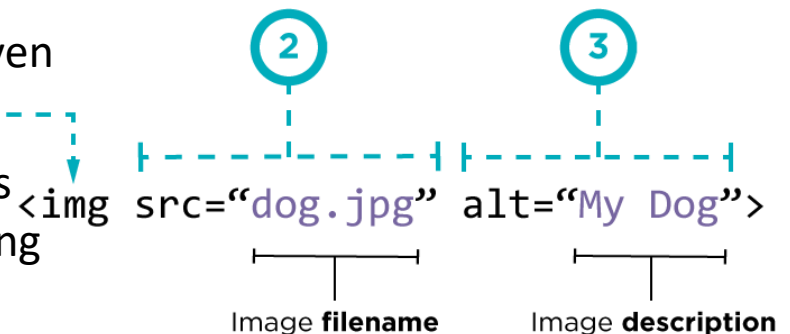


Image Element cont'd



Absolute vs. Relative URLs

- **Absolute links** are URLs that contain the protocol, the domain name and the full path to the resource
- **Relative links** are URLs that are 'relative' to the file that you are using them in, or linking from
- Relative path can be specified down the in the directory tree starting from the root directory

```
<body>
  <img src = "https://www.tutorialspoint.com/images/html.gif"
      alt = "HTML Tutorial" height = "250" width = "340" />
</body>

</html>
```

Image Element cont'd



The alt attributes:

- Defines an alternative text description of the image
- A browser might not display images for several reason, such as:
 - The image is invalid or an unsupported type or not available
 - The image cannot to be loaded because of slow connection
 - Non-visual browsers (such as those used by people with visual impairments)
 - The user chooses not to display images (saving bandwidth, privacy reasons)
- In these cases, the browser may replace the image with the text in the element's alt attribute
- Omitting alt indicates that the image is a key part of the content and no textual equivalent is available
- The alt attribute is considered good for SEO prospective
- Therefore, provide a useful value for alt attributes whenever possible

Image Element cont'd

- The **empty alt** attributes:

- Indicates that this image is **not a key** part of the content
- Non-visual browsers may omit it from rendering the image
- Visual browsers will also hide the broken image icon and the image failed to display

HTML image example with alt attribute



- Authoring meaningful alternate descriptions

- An alt attribute's value should clearly and concisely describe the image's content
- It should not describe the presence of the image itself or the file name of the image

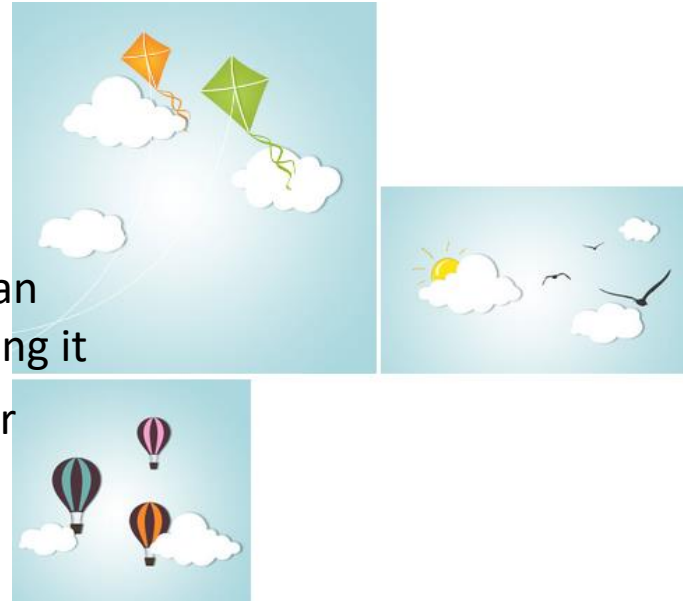
HTML image example with alt attribute

 animal image

Attribute	Value	Description
align	top, bottom, middle, left, right	<i>Deprecated</i> – Specifies the alignment for the image
alt	text	Specifies alternate text
border	pixels	<i>Deprecated</i> – Specifies the width of the image border.
crossorigin	anonymous use-credentials	It allows images from third-party sites that allow cross-origin access to be reused with canvas.
height	pixels or %	Specifies the height of the image.
hspace	pixels	<i>Deprecated</i> – Amount of white space to be inserted to the left and right of the object.
ismap	URL	Defines the image as a server-side image map.
longdesc	text	<i>Deprecated</i> – Specifies a URI/URL of a long description - this can elaborate on a shorter description specified with the alt attribute.
src	URL	the url of an image
usemap	#mapname	Defines the image as a client-side image map and used alongwith <map> and <area> tags.
vspace	pixels	<i>Deprecated</i> – Amount of white space to be inserted to the top and bottom of the object.
width	pixels or %	Sets the width of an image in pixels or in %.

Image Element cont'd

- It's a good practice to specify both the width and height attributes for an image, so that browser can allocate enough space for the image before loading it
- Otherwise, image loading may cause distortion or flicker in your website layout



```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Setting Image Width and Height Using style Attribute in
HTML</title>
</head>
<body>
  
  
  
</body>
</html>
```

Image Element cont'd

- Responsive Images the Simple Way (using sizes, src, srcset attributes)

1. At this screen size,

2. I will be displayed at roughly this size.

```

```

3. You can choose from any of these images...

4. ...which are this wide

5. So please pick the best image for right now.

Image Element cont'd



Supported image formats

- The image file formats that are most commonly used on the web are:
- APNG (Animated Portable Network Graphics)
 - Good choice for lossless animation sequences (GIF is less performant)
- AVIF (AV1 Image File Format)
 - Good choice for both images and animated images due to high performance
- GIF (Graphics Interchange Format)
 - Good choice for simple images and animations.
- JPEG (Joint Photographic Expert Group image)
 - Good choice for lossy compression of still images (currently the most popular)
- PNG (Portable Network Graphics)
 - Good choice for lossy compression of still images (slightly better quality than JPEG)
- SVG (Scalable Vector Graphics) - vector image format
 - Use for images that must be drawn accurately at different sizes.
- WebP (Web Picture format)
 - Excellent choice for both images and animated images

Creating Image Map

The image is a composite of three MDN Web Docs pages, each featuring a sidebar with navigation links and a main content area. A 3D pyramid is superimposed in the center, with its faces labeled 'CSS', 'HTML', and 'JS'. Arrows point from the pyramid's faces to the corresponding MDN page content.

- CSS: Cascading Style Sheets**
 - Table of contents
 - Key resources
 - Tutorials
 - Reference
 - Cookbook
 - Tools for CSS development
 - Meta bugs
 - See also
 - Cascading Style Sheets (CSS) is a [stylesheet](#) language used to describe the presentation of the content displayed on a web page, written in [HTML](#) or [XML](#) (including XML dialects such as [SVG](#), [MathML](#) or [XHTML](#)). Elements should be rendered on screen, on paper, in speech, or on other media.
 - CSS is among the core languages of the [open web](#) and is standardized by the [W3C](#) specifications. Previously, development of various parts of CSS was done asynchronously, which allowed versioning of the latest recommendations (CSS1, CSS2.1, CSS3). However, CSS4 has never become a recommendation.

- JavaScript**
- Table of contents
 - Tutorials
 - Reference
 - Tools & resources
- JavaScript (JS) is a lightweight, interpreted, or [just-in-time](#) compiled programming language with [first-class functions](#). While it is most well-known as the scripting language for Web pages, [many non-browser environments](#) also use it, such as [Node.js](#), [Apache CouchDB](#) and [Adobe Acrobat](#). JavaScript is a [prototype-based](#), multi-paradigm, single-threaded, dynamic language, supporting object-oriented, imperative, and declarative (e.g. functional programming) styles. Read more [about JavaScript](#).
- This section is dedicated to the JavaScript language itself, and not the parts that are specific to Web pages or other host environments. For information about [API](#) specifics to Web pages, please see [Web APIs](#) and [DOM](#).
- HTML: HyperText Markup Language**
- Table of contents
 - Key resources
 - Beginner's tutorials
 - Advanced topics
 - References
- HTML (HyperText Markup Language) is the most basic building block of the Web. It defines the meaning and structure of web content. Other technologies besides HTML are generally used to describe a web page's appearance/presentation ([CSS](#)) or functionality/behavior ([JavaScript](#)).

- An image map is an image with clickable areas

Creating Image Map cont'd



- Image map is defined by the <map> element with <area> elements
- The idea is to be able to perform different actions depending on where in the image you click
- It is an inline element and belongs to the phrasing content ----> Flow content model
- The <map> tag has one required attribute called name that gives the map a name so that it can be referenced
- The name attribute must:
 - Has a non-empty value with no space characters
 - Has a unique value – two map element in the same document not allowed to have the same name

Creating Image Map cont'd



- How Does it image map Works?
- The image is inserted using the `<img>` tag
 - Here the `<img>` element is different from other images because it require a usemap attribute
- Usemap attribute
 - Its value starts with a # followed by the name of the image map.
 - Used to create a relationship between the image and the image map
- Then, the `<map>` element is used to create an image map, and is linked to the image by using the required name attribute
- Then, the clickable areas are defined and added using an `<area>` element

Creating Image Map example

```
<h2>Image Maps</h2>
<p>Click on the computer, the phone, or the cup of coffee
    to go to a new page and read more about the topic:</p>



<map name="workmap">
  <area shape="rect" coords="34,44,270,350"
    alt="Computer" href="computer.htm">
  <area shape="rect" coords="290,172,333,250"
    alt="Phone" href="phone.htm">
  <area shape="circle" coords="337,300,44"
    alt="Cup of coffee" href="coffee.htm">
</map>
```

Creating Image Map cont'd



The <area> element

- It defines an area inside an image map that has predefined clickable areas
- Allows geometric areas on an image to be associated with hypertext link
- Used only within a <map> element
 - It is defined with (required) attributes shape and coords
 - Shape attribute - specifies the shape of the clickable area such as rectangle, circle, square, and polygon
 - Coords attribute - defines the coordinates of areas inside the image

Creating Image Map cont'd



How coordinate works?

- **Shape attribute** possible values
 - circle, rect, poly, default

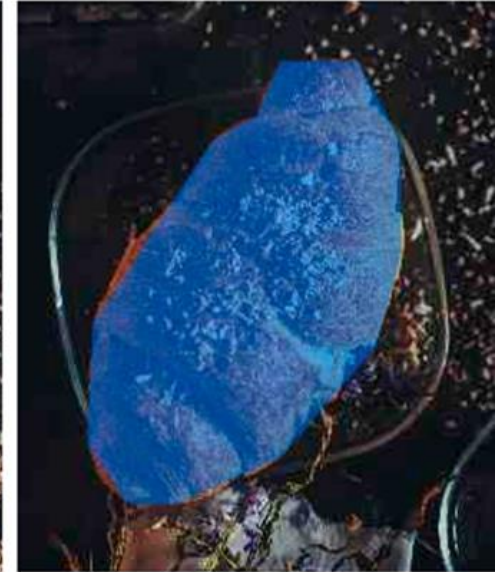
Cords attribute possible values

x, y, r (circle) where x, y specify the **center** and **r** is the radius of circle

Pair of x,y (rect) that specify the **left, top, right, bottom** coordinates of rectangle

x1,y1,x2,y2,x3,y3,..(polygon) that the polygon vertices

Note: *The easiest technique is using map coordinate generator*



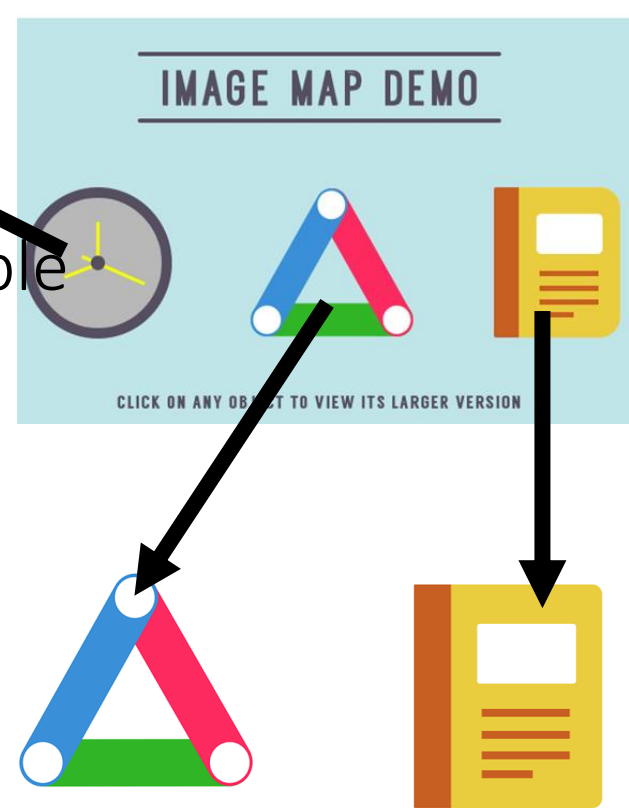
Creating Image Map cont'd



Attribute	Value(s)	Description
alt	Text	Alternative text to display if image can not be displayed
target	_self, _blank, _parent, _top	Specifies where to display the linked resource (current window, new window, the parent frame, full width in the original (same) window)
href	URL	Determine the hyperlink destination for the active (clickable) area
hreflang	language_code	• The language of the linked resource. • Must specified this attribute if you provide a value for the href attribute
download	File name	Indicates that the hyperlink will be used to download a resource
rel	alternate, author, help, next, bookmark, license, nofollow, noreferrer, prefetch, prev, search, tag	• Determine the relationship between the current document and the hyperlink • Used if a value for the href attribute is provided

Creating Image Map example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Creating Image Maps in HTML</title>
</head>
<body>
  
  <map name="objects">
    <area shape="circle" coords="137,231,71"
          href="/examples/html/clock.html" alt="Clock">
    <area shape="poly" coords="363,146,273,302,452,300"
          href="/examples/html/sign.html" alt="Sign">
    <area shape="rect" coords="520,160,641,302"
          href="/examples/html/book.html" alt="Book">
  </map>
</body>
</html>
```



The <picture> and <source> Element



The <picture>

- Unlike tag, it offer alternative versions of an image for different display/device scenarios
- Like tag, it is also an inline element and the content categories are embedded content ---> phrasing content ---> Flow content
- It nest a single element and zero or more <source> elements
- The element serves two purposes:
 - It describes the size and other attributes of the image and its presentation
 - It provides a fallback in case none of the offered <source> elements can provide a usable image
- Always specify the element as the last child element of <picture> element

The <picture> and <source> Element cont'd



- The <source> Element
 - It specifies multiple media resources for the <picture> element.
 - It also used with the <audio> element, and the <video> element.
 - It is an empty element (or void element) and also doesn't has content model
 - Commonly used to offer the same media content in multiple file formats in order to provide compatibility with a broad range of browser
- How <picture> and <source> elements work?
 - The browser will consider each child <source> element and choose the best match among them
 - If no matches are found or the browser doesn't support the <picture> element then the URL of the element specified in the src attribute is selected

The <picture> and <source> Element cont'd



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <meta name="viewport"
5       content="width=device-width, initial-scale=1.0">
6 </head>
7 <body>
8 <h2>The picture Element</h2>
9 <picture>
10   <source media="(min-width: 650px)"
11           srcset="https://www.w3schools.com/html/img_food.jpg">
12   <source media="(min-width: 465px)"
13           srcset="https://www.w3schools.com/html/img_car.jpg">
14   
16 </picture>
17 </body>
18 </html>
19
```

Considered If none of the source match the view point then

The <picture> and <source> Element cont'd



- The <source> attributes

Attribute	Value	Description
media	media_query	• media query of the resource's intended media • Accepts any valid media query that would normally be defined in a CSS
sizes	media condition-length pairs	• Specifies list of media source sizes for different page layouts • Each source size consists of a comma-separated list of media condition-length pairs. <i>E.g. sizes = "(max-height: 500px) 1000px"</i>
src	URL	Required when <source> is used in <audio> and <video> Specifies the URL of the media file
srcset	URL	• Required when <source> is used in <picture>. • Specifies the URL of the image to use in different situations
type	MIME-type	Specifies the MIME-type of the resource

The <picture> and <source> Element example



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
5 </head>
6 <body>
7
8 <h1>The picture element</h1>
9
10 <p>Resize the browser window to load different images.</p>
11
12 <picture>
13   <source media="(min-width:650px)"
  srcset="https://www.w3schools.com/tags/img_pink_flowers.jpg">
14   <source media="(min-width:465px)"
  srcset="https://www.w3schools.com/tags/img_white_flower.jpg">
15   
16 </picture>
17
18 </body>
```

Considered If none of the source match the view point then

The <picture> and <source> Element cont'd



Common use cases for <picture>

- Bandwidth
 - For small screen or device, it is not necessary to load a large image file.
 - By loading the most appropriate image for the viewer's display, the <picture> element allow saving bandwidth and speeding page load times .
- Format Support
 - Some browsers or devices may not support all image formats.
 - With the <picture> element, you can add images of all formats, and let the browser to find the supported format
- Art direction
 - Cropping or modifying images for different media conditions.
 - E.g. loading a simpler version of image which has too many details, on smaller displays

Audio and Video Elements



- Before HTML5 there hasn't been a standard multimedia on web
- Most videos were shown through a plugin. However, not all browsers have the same plugins
- HTML5 features include native audio and video support without the need for Flash/a plugin
- A standard way to include multimedia contents (video or audio) on web
 - The <audio> tag - embed sound content in web documents
 - The <video> tag - embed a media player which supports video playback
 - The <source> tag – specify the source of media both for audio and video
- Both <audio> and <video> tags require the media controls

Audio and Video Elements cont'd



- **Media Sources**

One or more media sources are represented using the **src attribute** and the **<source> element**

- **Examples:**

```
<!DOCTYPE html>
<html>
<body>

<audio controls>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
  Your browser does not support the audio element.
</audio>

</body>
</html>
```

```
<!DOCTYPE html>
<html>
<body>

<video width="320" height="240" controls>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
  Your browser does not support the video tag.
</video>

</body>
</html>
```

Attribute	Value	Description
autoplay	autoplay	A Boolean attribute: if specified, the audio/video will automatically begin playback as soon as it is ready
controls	controls	if specified, the controls displayed to allow the user to control audio/video playback, including volume, seeking, and pause/resume.
loop	loop	Audio/video will start over again, every time it is finished
muted	muted	A Boolean attribute that indicates whether the audio will be initially silenced. Its default value is false.
preload	non / auto / metadata	Enumerated attribute which intended to provide a hint to the browser what content is loaded before the media played.
height	pixels	<i>Sets the height of the video player</i>
width	pixels	<i>Sets the width of the video player</i>
poster	URL	<i>An image to be shown while the video is downloading, or until the user hits the play button</i>

HTML Audio and Video Formats



- There are three supported video and audio formats.
- The browser support for the different formats is given below:

Browser	Video format			Audio format		
	MP4	WebM	Ogg	MP3	WAV	OGG
Edge	YES	YES	YES	YES	YES*	YES*
Chrome	YES	YES	YES	YES	YES	YES
Firefox	YES	YES	YES	YES	YES	YES
Safari	YES	YES	NO	YES	YES	NO
Opera	YES	YES	YES	YES	YES	YES

Audio and Video Elements

Video example



```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Embedding Video into an HTML Page</title>
</head>
<body>
  <video controls="controls" src="/examples/video/shuttle.mp4">
    Your browser does not support the HTML5 Video element.
  </video>

  <p> A video with alternative sources. </p>
  <video controls="controls">
    <source src="/examples/video/shuttle.mp4" type="video/mp4">
    <source src="/examples/video/shuttle.ogv" type="video/ogg">
    Your browser does not support the HTML5 Video element.
  </video>
</body>
</html>
```

Audio and Video Elements

Audio example



```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Specify Alternate Sources for audio Element in
HTML</title>
</head>
<body>
  <p> A audio with single sources. </p>
  <audio controls="controls" src="/examples/audio/birds.mp3">
    Your browser does not support the HTML5 audio element.
  </audio>
  <br><br/> <hr/>
  <p> A audio with alternative sources. </p>
  <audio controls="controls">
    <source src="/examples/audio/birds.mp3" type="audio/mpeg">
    <source src="/examples/audio/birds.ogg" type="audio/ogg">
    Your browser does not support the HTML5 audio element.
  </audio>
</body>
</html>
```

The <track> Element

- HTML <track> tag is used to define time-based text tracks for a media file.
- It must be used as child element of <audio> and <video> elements.
- It is used to specify subtitles, caption files or other files containing text, that should be visible when the media is playing.
- Tracks are formatted in WebVTT format (.vtt files).

The <track> Element attributes



Attribute	Value	Description
default	default	It specifies that the track should be enabled unless the user's preferences indicate that another track is more important
kind	captions / chapters descriptions / metadata subtitles	It specifies that which type of text track you want to add
label	text	It specifies the title of the text track
src	URL	It defines the URL of the track file
srclang	language_code	It defines the language of the track text content, such as English, Germany, etc.

The <track> Element cont'd



How to create WEBVTT file:

- In the text editor (notepad)
 1. Write *WEBVTT* as the first line in the editor
 2. Leave a blank line
 3. Specify the time duration in the proper format
00:01.000 --> 00:04.000
 4. Enter and write your text which you want to add a subtitle or caption
- Never drink liquid nitrogen.
 5. Repeat step 3 to 5 until you finish it
 6. Save it using .vtt extension

Example 1:

WEBVTT - This file has cues.

00:01:14.815 --> 00:01:18.114

- What?

- Where are we now?

00:01:18.171 --> 00:01:20.991

- This is big bat country.

00:01:21.058 --> 00:01:23.868

- [Bats Screeching]

- They won't get in your hair.

- They're after the bugs

Note: For more details checkout

- https://developer.mozilla.org/en-US/docs/Web/API/WebVTT_API

The <figure> and <figurecaption> Elements



- The <figure> element
 - Specifies self-contained content, like illustrations, diagrams, photos, code listings, poem script etc
 - Its content is related to the main flow, but its position is independent of the main flow, and if removed it should not affect the flow of the document
- The <figurecaption> element
 - Represents a caption or legend describing the contents of <figure> element which is its parent
 - Can be placed as the first or last child of the <figure> element
- Both <figure> tag and <figurecaption> tag don't have any specific attribute

The <figure> and <figurecaption> Elements cont'd



Example 1:

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Example of HTML figure Tag</title>
</head>
<body>
  <figure>
    
    <figcaption>An elephant at sunset</figcaption>
  </figure>
</body>
</html>
```



An elephant at sunset

Note The *figure*, its *caption*, and its *contents* are referenced as a **single unit**

The <figure> and <figurecaption> Elements cont'd



Output

Example 2:

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Example of HTML figure Tag</title>
</head>
<body>
  <figure>
    <p style="white-space:pre">
Bid me discourse, I will enchant thine ear
  Or like a fairy trip upon the green,
Or, like a nymph, with long dishevell'd
Dance on the sands, and yet no footing seen:
Love is a spirit all compact of fire,
Not gross to sink, but light, and will
    <figcaption><cite>Venus and Adonis</cite>
      by William Shakespeare</figcaption>
  </figure>
</body>
</html>
```

Bid me discourse, I will enchant thine ear,
Or like a fairy trip upon the green,
Or, like a nymph, with long dishevell'd hair,
Dance on the sands, and yet no footing seen:
Love is a spirit all compact of fire,
Not gross to sink, but light, and will aspire.

Venus and Adonis, by William Shakespeare

Note:

Only one <figcaption> element can be nested within a <figure> tag although the <figure> element itself may contain multiple other elements like or <code>

HTML Table Element



- Tables play an essential role in organizing complex data in a simple and logical way, and presenting it in easy follow
- The HTML table model allows authors to arrange data (text, preformatted text, images, links, forms, form fields, other tables, etc.) in tabular
- Before taking a look how to use tables in HTML, let's take a look at *Table structure*

Table caption → **Table 1: Course Price**

Table header		
Title	Price	Available
CSS Demystified	\$29	16
Mastering JavaScript	\$35	10
HTML5: An Introduction	\$15	6
Total		32

Table rows

Table cell

Table body

Table footer

Table columns

HTML Table Element cont'd



Element	tag	Description
Table	<code><table></code> <code></table></code>	• Defines the beginning and end of the table • A container tag that holds the other components of the html table
Table rows	<code><tr></code> <code></tr></code>	• Defines the beginning and end of a horizontal row • May be grouped into a head, foot, and body sections
Table cells	<code><td></code> <code></td></code>	• Defines an individual cell • Cells are always placed inside a row

HTML Table Element cont'd

Example 1: Creating a table with default style

```
<table>
<tr>
  <td> Title </td>
  <td> Price </td>
  <td> Available </td>
</tr>
<tr>
  <td> CSS Demystified </td>
  <td> $29 </td>
  <td> 16 </td>
</tr>
<tr>
  <td> Mastering JavaScript </td>
  <td> $35 </td>
  <td> 10 </td>
</tr>
<tr>
  <td> HTML5: An Introduction </td>
  <td> $15 </td>
  <td> 6 </td>
</tr>
</table>
```

Title	Price	Available
CSS Demystified	\$29	16
Mastering JavaScript	\$35	10
HTML5: An Introduction	\$15	6

Note:

By default table doesn't have borders

HTML Table Element cont'd



Element	tag	Description
Table caption	<code><caption> </caption></code>	• Specifies the caption (or title) of a table that provides a short description of the table's purpose • Goes after the table tag but not inside “tr” or “td.” • Only single caption per table • Appears centered above the table
Table header	<code><th> </th></code>	• Defines a table header row. Used instead of “td.” • Text is bold & centered within the cell
Table section	<code><thead> </thead></code>	Defines a set of rows defining the head of the columns of the table
	<code><tbody> </tbody></code>	Encapsulates a set of table rows (<tr> elements), indicating that they comprise the body of the table
	<code><tfoot> </tfoot></code>	Defines a set of rows summarizing the columns of the table

HTML Table Element example

```

<table>
  <caption> Course Price</caption>
  <thead>
    <tr><th> Title </th> <th> Price </th> <th> Available </th> </tr>
  </thead>

  <tbody>
    <tr><td> CSS Demystified </td> <td> $29 </td> <td> 16 </td> </tr>
    <tr><td> Mastering JavaScript</td><td> $35 </td> <td> 10 </td> </tr>
    <tr><td> HTML5: An Introduction </td><td> $15 </td><td> 6 </td></tr>
  </tbody>

  <tfoot>
    <tr><td> Total </td> <td></td> <td> 32 </td> </tr>
  </tfoot>
</table>
  
```

Title	Price	Available
CSS Demystified	\$29	16
Mastering JavaScript	\$35	10
HTML5: An Introduction	\$15	6
Total		32

HTML Table Element attributes



- All most all attributes of the table elements are deprecated.
- Only *<td>* and *<tr>* tags have some attributes described below
<td> tag and <tr> tag attributes

Attribute	Value	Description
colspan	<i>Positive integer <=1000</i>	• Indicates for how many columns the cell extends (span) • Its default value is 1
rowspan	<i>Positive integer <=65534</i>	• Indicates for how many rows the cell extends (span). • Its default value is 1; if its value is set to 0, it extends until the end of the table section
headers	<i>Header_id</i>	Specifies one or more header cells a table cell is related to
scope	<i>row / col / rowgroup / colgroup</i>	<i>Enumerated attribute defines the cells that the header (defined in the <th>) element relates to</i>

HTML Table Element cont'd

- **Example:** Merging Cells with rowspan and colspan attributes

```
<table>
  <thead>
    <tr>
      <th rowspan = "2">Browser</th>
      <th colspan = "2">Vistor</th>
    </tr>
    <tr>
      <th>Number</th>
      <th>Precentage</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Mozila Firefox</td><td>163</td><td>59%</td> </tr>
      <tr><td>Google Chrome</td><td>78</td> <td>28%</td> </tr>
      <tr><td>Safari</td><td>35</td><td>13%</td> </tr>
    </tbody>
  </table>
```

Browser	Visitors	
	Number	Percentage
Mozilla Firefox	163	59%
Google Chrome	78	28%
Safari	35	13%

HTML Table Element cont'd



- The `<colgroup>` element
 - It defines a group of columns within a table for formatting .
 - It is useful for applying styles to entire columns, instead of repeating the styles for each cell, for each row.
 - To define different properties to a column within a `<colgroup>` element, it require the `<col>` tag
- The `<col>` element
 - It specifies column properties (common semantics) for each column (all common cells) within a `<colgroup>` element.
 - The common semantic (like color, padding etc.) is specified using CSS

HTML Table Element cont'd



- Both elements (<colgroup> and <col>) has single attribute named “span”
 - which is a positive integer indicating the number of consecutive columns the elements span

Example :

	Section A		Section B	
Class	Male	Female	Male	Female
8	30	40	30	20
9	20	30	30	20
10	10	20	20	30
Total	60	90	80	70

HTML Table Element cont'd



The colgroup element

Example – using CSS:

ISBN	Title	Price
3476896	My first HTML	\$53
5869207	My first CSS	\$49

```
<h3>The colgroup element</h3>
<table>
  <colgroup>
    <col span="2" style="background-color:red">
    <col style="background-color:yellow">
  </colgroup>
  <tr><th>ISBN</th><th>Title</th><th>Price</th></tr>
  <tr>
    <td>3476896</td><td>My first HTML</td><td>$53</td>
  </tr>
  <tr><td>5869207</td><td>My first CSS</td><td>$49</td>
  </tr>
</table>
```



Web Development Using HTML

Internet Programming I: Chapter 2 – Part III



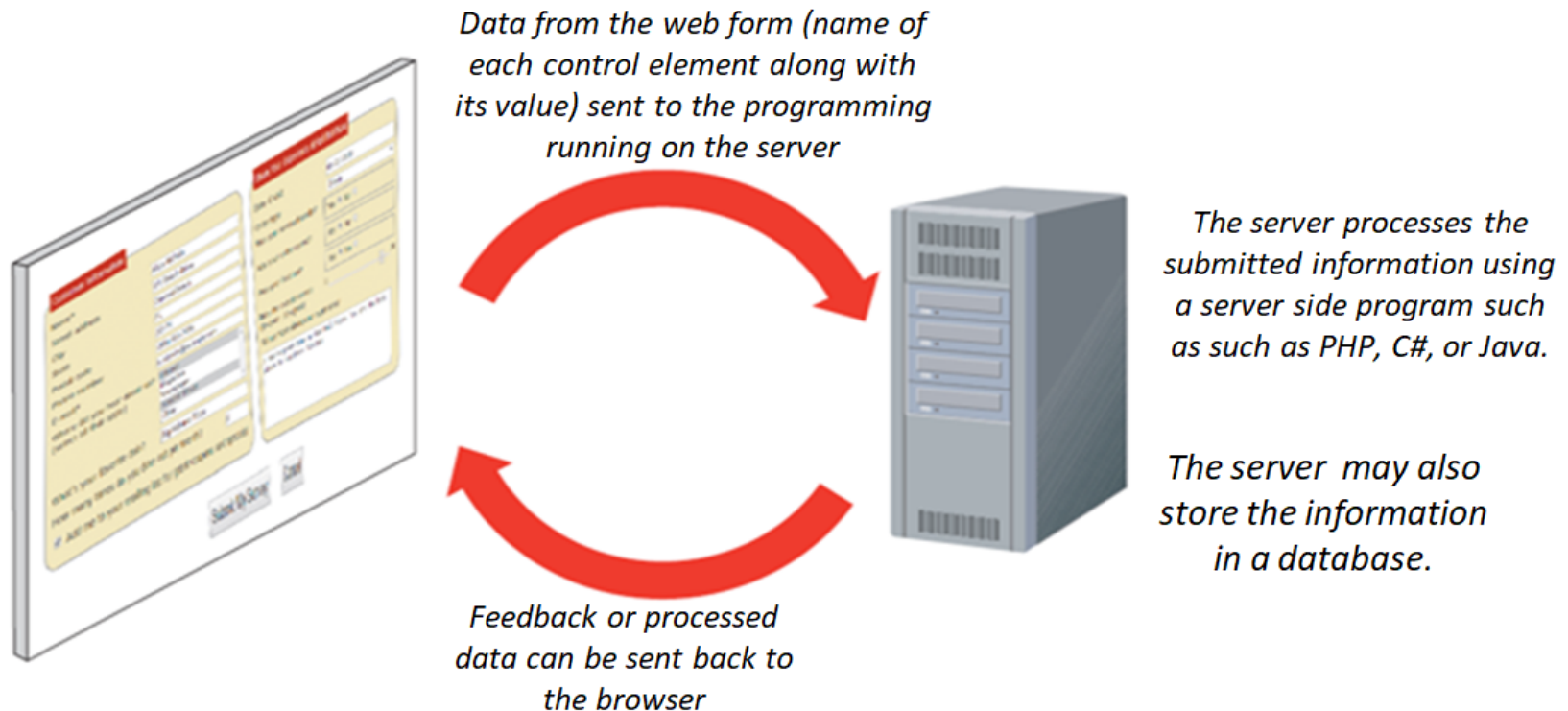
ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY
Department of Software Engineering

Working with Web Form



- Web form:
 - Allows users to enter data that can be saved and processed.
 - Common way to accept user input (collect some data from the site visitor)
 - E.g. user registration – capture user info like name, address, credit card, etc.
 - It will post the data captured from user to a back-end application such as CGI, ASP Script or PHP script etc.
 - The backend app will perform required processing on the passed data based on defined business logic inside the application
 - Allows the creation of interactive websites for user feedback
- The `<form>` HTML element :
 - Represents a web document section containing interactive controls for collecting and submitting information

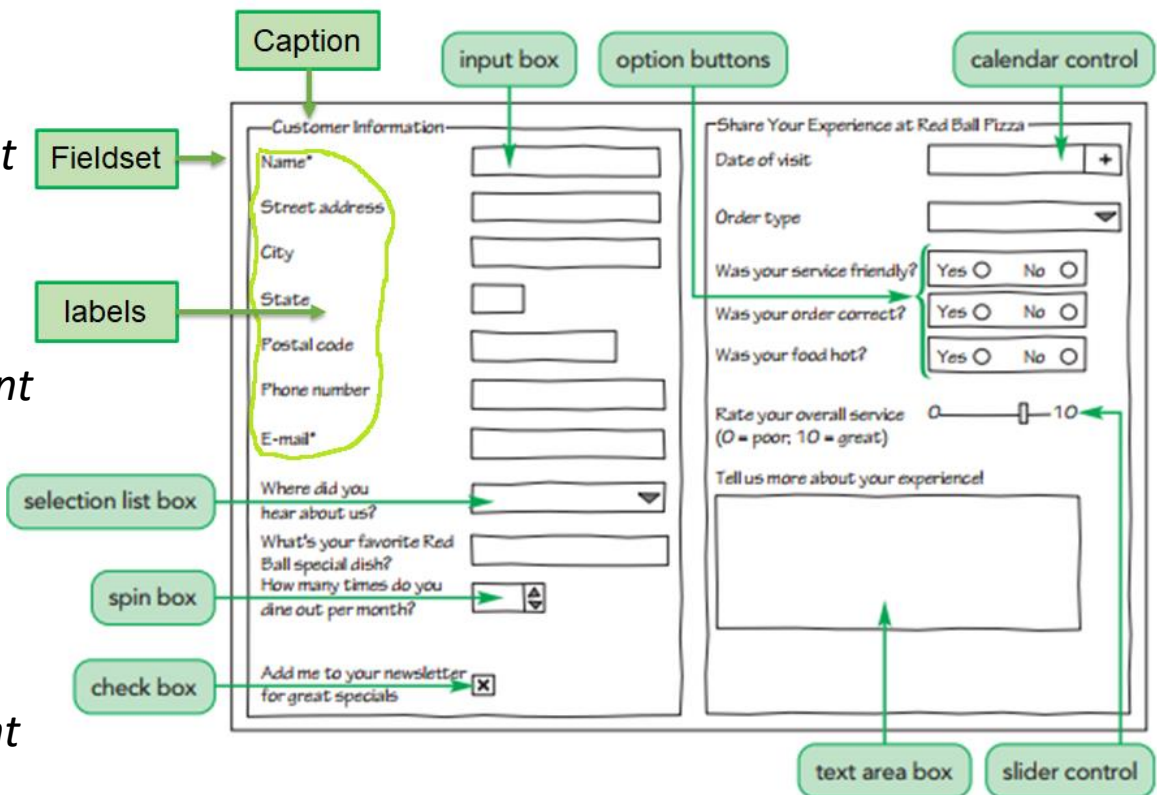
Working with Web Form cont'd



Working with Web Form cont'd

- The web form can contain one or more of the following

- `<input>` element
- `<textarea>` element
- `<button>` element
- `<select>` element
- `<option>` element
- `<optgroup>` element
- `<fieldset>` element
- `<legend>` element
- `<label>` element
- `<output>` element
- `<datalist>` element
- `<progress>` element
- `<meter>` element



The diagram illustrates a web form with two main sections. The first section, 'Customer Information', is enclosed in a fieldset and contains several input fields: Name*, Street address, City, State, Postal code, Phone number, and E-mail*. The second section, 'Share Your Experience at Red Ball Pizza', includes a date picker for 'Date of visit', a dropdown for 'Order type', three sets of radio buttons for service quality (friendly, order correct, food hot), a slider for 'Rate your overall service' (0 to 10), and a large text area for 'Tell us more about your experience!'. A checkbox is also present for 'Add me to your newsletter for great specials'. Various callouts point to specific controls: 'Caption' points to the form title, 'input box' points to a text field, 'option buttons' points to the radio buttons, 'calendar control' points to the date picker, 'Fieldset' points to the 'Customer Information' container, 'labels' points to the form labels, 'selection list box' points to the 'Order type' dropdown, 'spin box' points to the 'How many times do you dine out per month?' spinner, 'check box' points to the newsletter checkbox, 'text area box' points to the experience feedback text area, and 'slider control' points to the overall service rating slider.

From control elements



- These are the objects that allow a user to interact with a form.
- Each *data entry control element* is associated with a **data field** that stores the data values supplied by a user.
- Types of **controls**
 - **Text Input boxes**
 - Single line input
 - Multi-line input
 - Password input
 - **Choice/Selection**
 - Selection lists
 - Radio buttons
 - Check boxes
 - **Widget elements**
 - Spin boxes
 - Slider controls
 - Calendar controls
 - Color pickers

Creating Web forms



- Web forms are marked using the form element
 <form id="text" attributes>
 form control elements goes here
 </form>
 - id attribute used to uniquely identifies the form
 - attributes specify additional attributes of the form
 - Some of the form attributes control the behavior during form submission
- A form element can be placed anywhere within the body of a page
- Forms also can contain other web page elements such as tables, paragraphs, inline images, and headings

Creating Web forms example



```
<html>
<body>

<h1>The fieldset element</h1>

<form id = "registration" action=" " method="get">
  <fieldset>
    <legend>Personalia:</legend>
    <label for="fname">First name:</label>
    <input type="text" id="fname" name="fname"><br><br>
    <label for="lname">Last name:</label>
    <input type="text" id="lname" name="lname"><br><br>
    <label for="email">Email:</label>
    <input type="email" id="email" name="email"><br><br>
    <label for="birthday">Birthday:</label>
    <input type="date" id="birthday" name="birthday"><br><br>
    <input type="submit" value="Submit">
  </fieldset>
</form>

</body>
</html>
```

The Form element

Personalia: _____

First name:

Last name:

Email:

Birthday: 

Most common form attributes



Attribute	Value	Description
action	<i>URL</i>	Specifies where to send the form-data when a form is submitted (provide the location of web server program)
method	get / post	Specifies the HTTP method to use when sending form-data
novalidate	Novalidate	Specifies that the form should not be validated when submitted
accept-charset	<i>character_set</i>	Specifies the character encodings that are to be used for the form submission
autocomplete	<i>On / off</i>	<i>Specifies whether a form should have autocomplete or not</i>
name	String	<i>The name of the form. The value must be unique among the form elements and also must not be the empty</i>
target	<i>_blank / _self / _parent / _top</i>	<i>Specifies where to display the response that is received after submitting the form</i>

GET vs POST methods



GET	POST
Only limited amount of data can be sent because data is sent in header.	Large amount of data can be sent because data is sent in body.
Get request is not secured because query string appended in the URL bar.	Post request is secured because data is not exposed in the URL bar.
Get request can be bookmarked	Post request cannot be bookmarked.
A Get request is often cacheable.	A Post request can hardly cacheable.
Get request is more efficient and used more than post.	Post request is less efficient and used less than Get.

Grouping Form Control Elements



- **Field set:**
 - Groups fields (control elements) that share a common purpose
 - Field sets are created using the **fieldset element**
- **Legend:**
 - Describes the content of a field set using the **legend element**
 - Contains only text and no nested elements
 - By default, it placed in the top-left corner of the field set box and can be moved to a different location using the CSS positioning styles



Form Input Element



- The `<input type=" ">` is an important element of HTML form.
- It is the most commonly used element to create interactive controls for web-based forms in order to accept data from the user.
- A wide variety of the types of input data and control widgets are available, depending on the device and user agent.
- Syntax:

`<input name="name" id="id" type="type" />`

- Name attribute – provides the name of the data field associated with the control
- id attribute – uniquely identifies the control in which the user enters the value
- type attribute - indicates the input type (data type) of the field

HTML Input Types



- The "**type**" attribute of input element can be various types, which defines data field
- List the types of <input> element given in the table

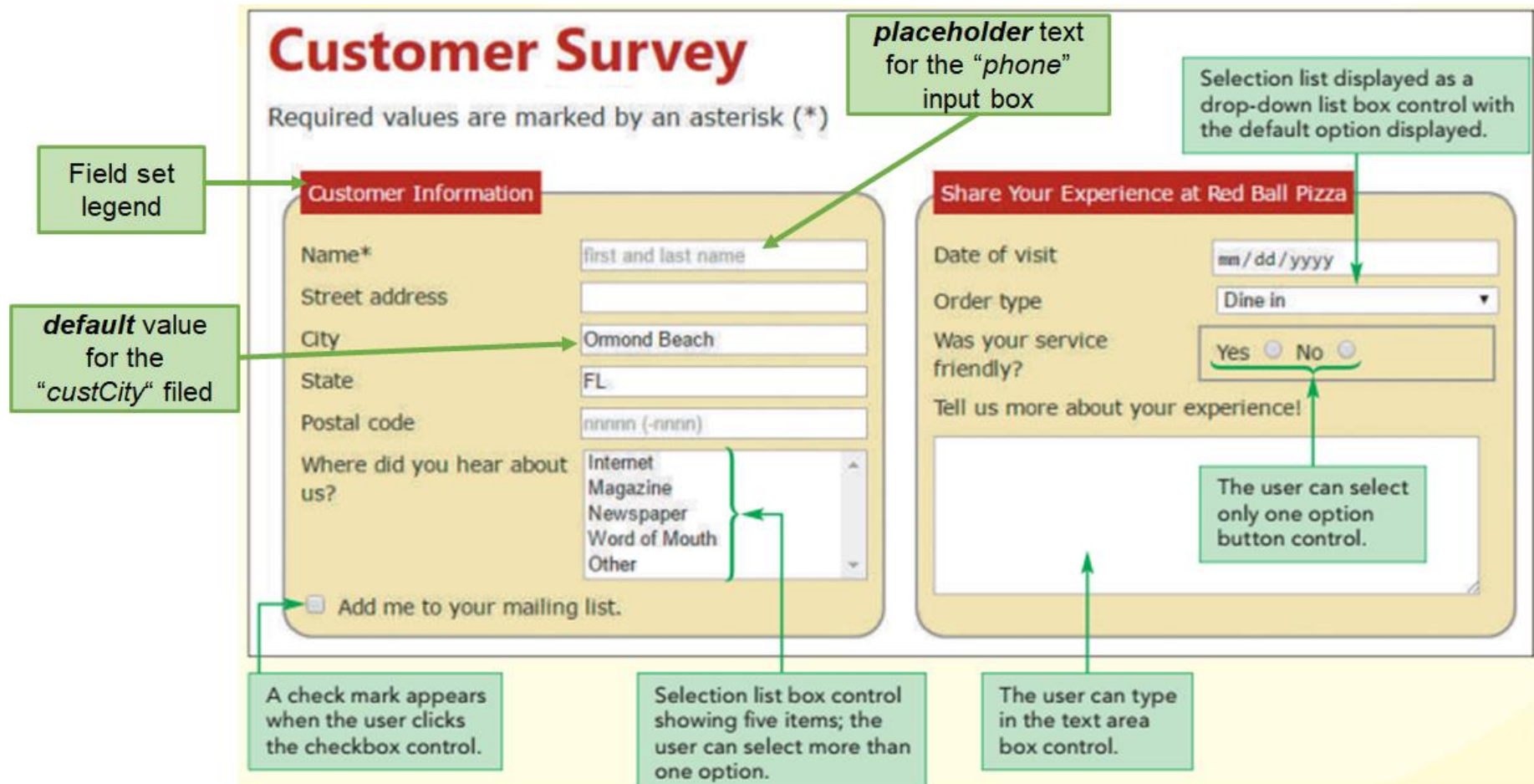
type=" "	Description
text	Defines a one-line text input field
password	Defines a one-line password input field
radio	Defines a radio button which allows select one option
checkbox	Defines checkboxes which allow select multiple options form.
file	Defines to select the file from device storage.
submit	Defines a submit button to submit the form to server.
reset	Defines a reset button to reset all values in the form.
button	Defines a simple push button, which programmed to perform a task on an event
image	Defines a graphical submit button.

HTML5 added new types on <input> element



type=" "	Description
color	Defines an input field with a specific color.
date	Defines an input field for selection of date.
datetime-local	Defines an input field for entering a date without time zone
email	Defines an input field for entering an email address.
month	Defines a control with month and year, without time zone.
number	Defines an input field to enter a number.
url	Defines a field for entering URL
week	Defines a field to enter the date with week-year, without time zone.
search	Defines a single line text field for entering a search string
tel	Defines an input field for entering the telephone number

Design and layout of the survey form example



Customer Survey

Required values are marked by an asterisk (*)

Customer Information

Name*

Street address

City

State

Postal code

Where did you hear about us?

- Internet
- Magazine
- Newspaper
- Word of Mouth
- Other

☐ Add me to your mailing list.

Share Your Experience at Red Ball Pizza

Date of visit

Order type

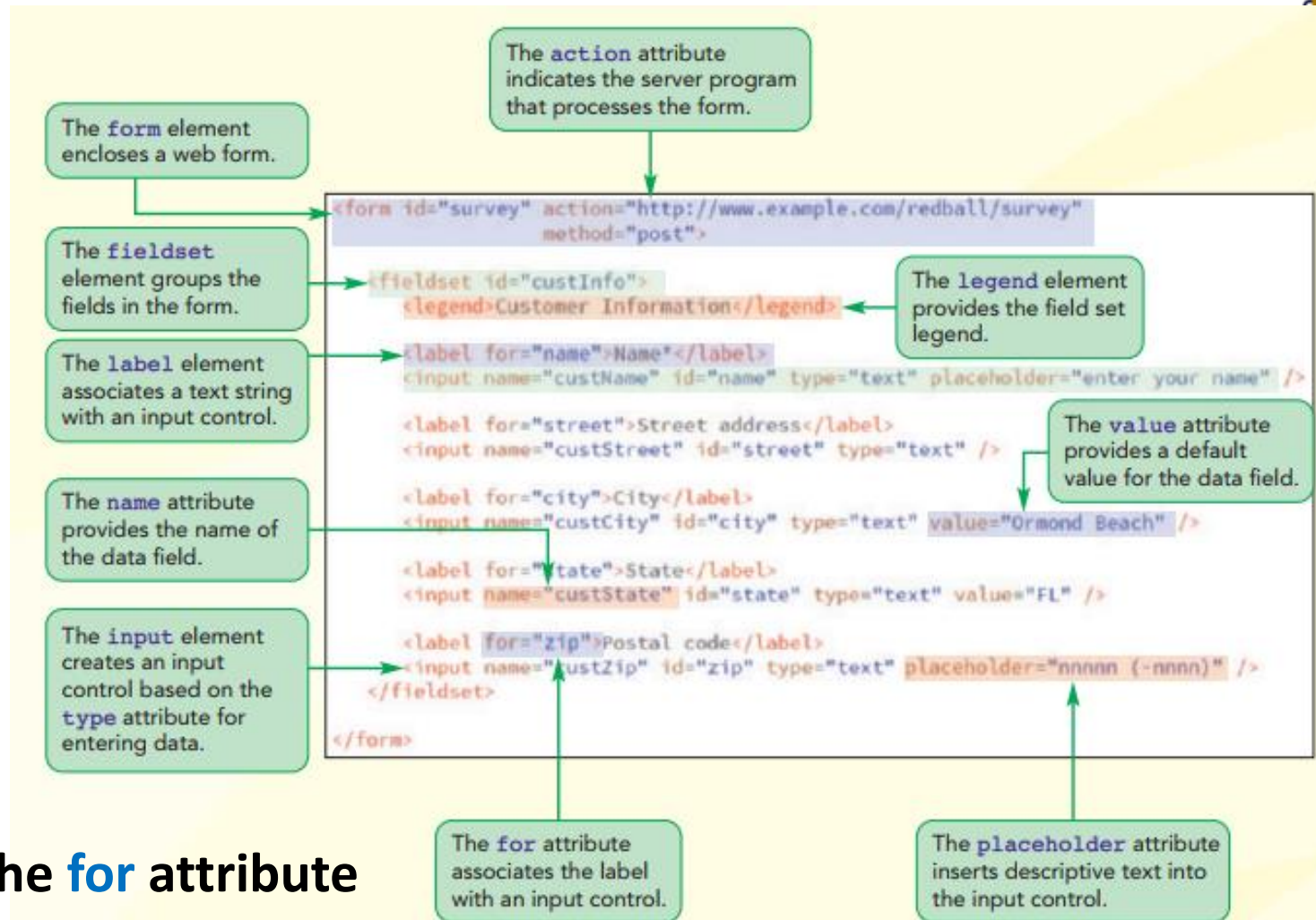
Was your service friendly? Yes ☐ No ☐

Tell us more about your experience!

Annotations:

- Field set legend
- placeholder text for the "phone" input box
- Selection list displayed as a drop-down list box control with the default option displayed.
- default value for the "custCity" filed
- A check mark appears when the user clicks the checkbox control.
- Selection list box control showing five items; the user can select more than one option.
- The user can type in the text area box control.
- The user can select only one option button control.

Design and layout of the survey form example



- **The `for` attribute**

- When used with the `<label>` elements specifies which form element a label is bound to

Design and layout of the survey form example

The `select` element creates a drop-down list box control.

Each option in a selection list is marked with the `option` element.

The `selected` attribute identifies the default option in a selection list.

Each radio button within an option group belongs to the same data field.

The `textarea` element marks a text area box control.

The `size` attribute sets the number of visible options.

The `multiple` attribute allows for multiple selections from the drop-down list.

The `checkbox` data type creates a checkbox control.

The `radio` data type creates an option button control.

```
<label for="infoSrc">Where did you hear about us?</label>
<select name="infoSrc" id="info" size="5" multiple>
  <option value="internet">Internet</option>
  <option value="mag">Magazine</option>
  <option value="news">Newspaper</option>
  <option value="word">Word of Mouth</option>
  <option value="other">Other</option>
</select>

<input name="mailMe" id="mailCB" value="yes" type="checkbox" />
<label for="mailCB">Add me to your mailing list.</label>

<label for="orderType">Order type</label>
<select name="orderType" id="order">
  <option value="order1">Carry out</option>
  <option value="order2">Delivery</option>
  <option value="order3" selected>Dine in</option>
  <option value="order4">Take 'n bake</option>
</select>

<label>Was your service friendly?</label>
<fieldset class="optGroup">
  <label for="fYes">Yes</label>
  <input name="sFriend" id="fYes" value="yes" type="radio" />
  <label for="fNo">No</label>
  <input name="sFriend" id="fNo" value="no" type="radio" />
</fieldset>

<label for="commBox">Tell us more about your experience!</label>
<textarea name="custExp" id="commBox"></textarea>
```

Common Attributes to all Input Types



Attribute	Description
autocomplete	Hint for form autofill feature
autofocus	Automatically focus the form control when the page is loaded
disabled	Define whether the form control is disabled
form	Associates the control with a form element
name	Name of the form control. Submitted with the form as part of a name/value pair.
value	The initial value of the control
readonly	Boolean. The value is not editable
required	Boolean. A value is required or must be check for the form to be submittable

Form Attributes



Attribute	Description
<code>action</code>	URL to use for form submission
<code>enctype</code>	Form data set encoding type to use for form submission
<code>method</code>	HTTP method to use for form submission
<code>novalidate</code>	Bypass form control validation for form submission

Other Attributes



Attribute		Description
numeric types	<u>max, min</u>	Maximum and Minimum value respectively
numeric types	<u>step</u>	Incremental values that are valid.
password, search, tel, text, url	<u>maxlength, minlength</u>	Maximum and Minimum length (number of characters) of value respectively
password, search, tel, text, url	<u>placeholder</u>	Text that appears in the form control when it has no value set
email, password, tel, text, url	<u>size</u>	Size of the control
password, text, tel	<u>pattern</u>	Pattern the value must match to be valid
email, file	<u>multiple</u>	Boolean. Whether to allow multiple values
file	<u>accept</u>	Hint for expected file type in file upload controls

Designing a Form Layout

- There are two general layouts
 - *Labels are placed directly above the input controls*
 - *Labels and controls are placed side-by-side*

Customer Information

Name *

Street address

City

State

one-column layout

Customer Information

Name *

Street address

City

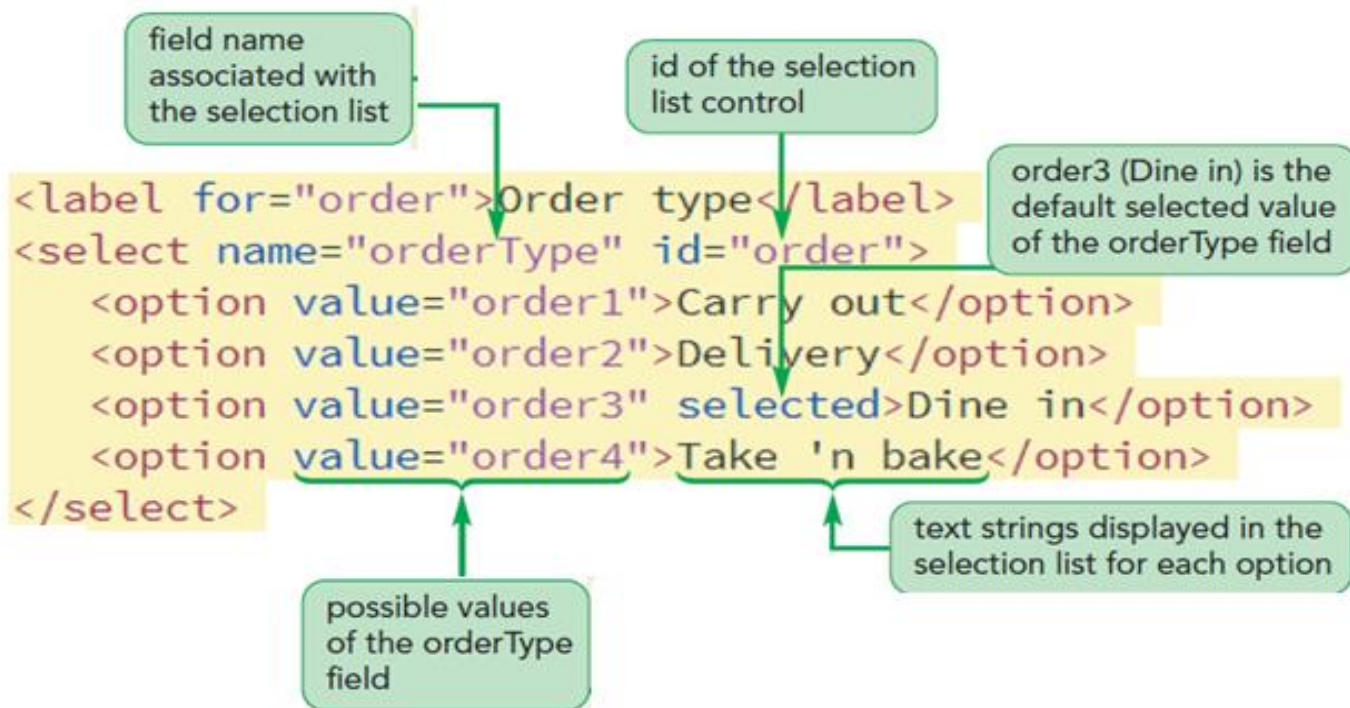
State (abbr.)

two-column layout

Creating a Selection List

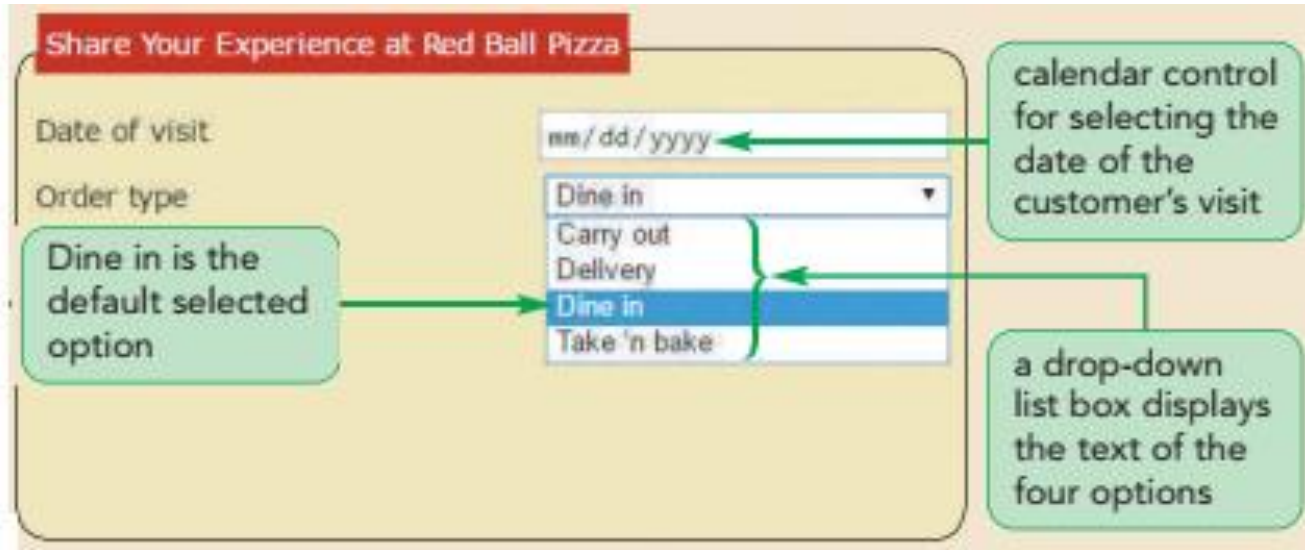


- A selection list is a list box that presents users with a group of possible values for the data field
- The list is created using the select and option elements



Multiple items selection

- Two ways for users to select multiple items from a selection list
 - For non-contiguous selection, press and hold the Ctrl key while making the selections
 - For contiguous selection, select the first item, press and hold the Shift key, and then select the last item in the range



The screenshot shows a web form with the title "Share Your Experience at Red Ball Pizza". It contains two main input fields: "Date of visit" and "Order type".

The "Date of visit" field is a text input with a placeholder "mm/dd/yyyy". A green callout box points to it with the text "calendar control for selecting the date of the customer's visit".

The "Order type" field is a drop-down list box. It has four options: "Dine in", "Carry out", "Delivery", and "Take 'n bake". The "Dine in" option is selected and highlighted in blue. A green callout box points to the list with the text "a drop-down list box displays the text of the four options". Another green callout box points to the "Dine in" option with the text "Dine in is the default selected option".

- *By default, a selection list appears as a drop-down list box*
- *To display a selection list as a scroll box, use the size attribute*

Grouping Selection Options

- The selection list options can be organized by placing them in option groups using the **optgroup** element

```
<label for="appetizers">Starter Menu</label>
<select name="meal">
  <optgroup label="Appetizers">
    <option value="sms">Spicy Mozzarella Sticks</option>
    <option value="pr">Pepperoni Rolls</option>
    <option value="tr">Toasted Ravioli</option>
  </optgroup>
  <optgroup label="Salads">
    <option value="sms">Pasta Salad</option>
    <option value="tbs">Tuscan Bread Salad</option>
    <option value="pr">Caesar Salad</option>
  </optgroup>
</select>
```

option group labels



Spicy Mozzarella Sticks ▼
Appetizers
Spicy Mozzarella Sticks
Pepperoni Rolls
Toasted Ravioli
Salads
Pasta Salad
Tuscan Bread Salad
Caesar Salad

Data Lists - suggesting options



- Data list is a list of possible data values that a form field can have
- It defined using the datalist element

```
<label for="dish">What's your favorite dish?</label>
<input name="favDish" id="dish" type="text" list="dishType" />
<datalist id="dishType">
  <option value="Anitpasto Pizza" />
  <option value="Big Kahuna Pizza" />
  <option value="BBQ Chicken Pizza" />
  <option value="Mediterranean Herb Pizza" />
  <option value="Pasta Rolls" />
  <option value="Pasto Artichoke Pizza" />
</datalist>
```

data list containing suggested values

links the favDish field to the dishType data list

A screenshot of a web form with three sections. The first section is a multi-select dropdown labeled "Where did you hear about us? (select all that apply)" with options: Internet, Magazine, Newspaper, Word of Mouth, and Other. The second section is a text input field labeled "What's your favorite dish?" with the letter "p" entered. Below this field, a dropdown menu is open, showing "Pasta Rolls" and "Pasto Artichoke Pizza" as suggested options. A green arrow points from the "Pasta Rolls" option in the dropdown to a green callout box that says "suggested values from the data list starting with the letter 'p'". The third section is a checkbox labeled "Add me to your mailing list".

Entering Date and Time Values



- Date and time fields ensure that users enter data in the correct format
- Indicated using type attributes: date, time, datetime-local, month, and week

August 2021

August 2021

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31	1	2	3	4

type="month"

Week 33, 2021

August 2021

Week	Sun	Mon	Tue	Wed	Thu	Fri	Sat
31	1	2	3	4	5	6	7
32	8	9	10	11	12	13	14
33	15	16	17	18	19	20	21
34	22	23	24	25	26	27	28
35	29	30	31	1	2	3	4

type="week"

08/17/2021

August 2021

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31	1	2	3	4

type="date"

08/17/2021 04:21 PM

August 2021

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31	1	2	3	4

type="datetime-local"

```
<label for="visit">Date of visit</label>  
<input name="visitDate" id="visit" type="date" />
```

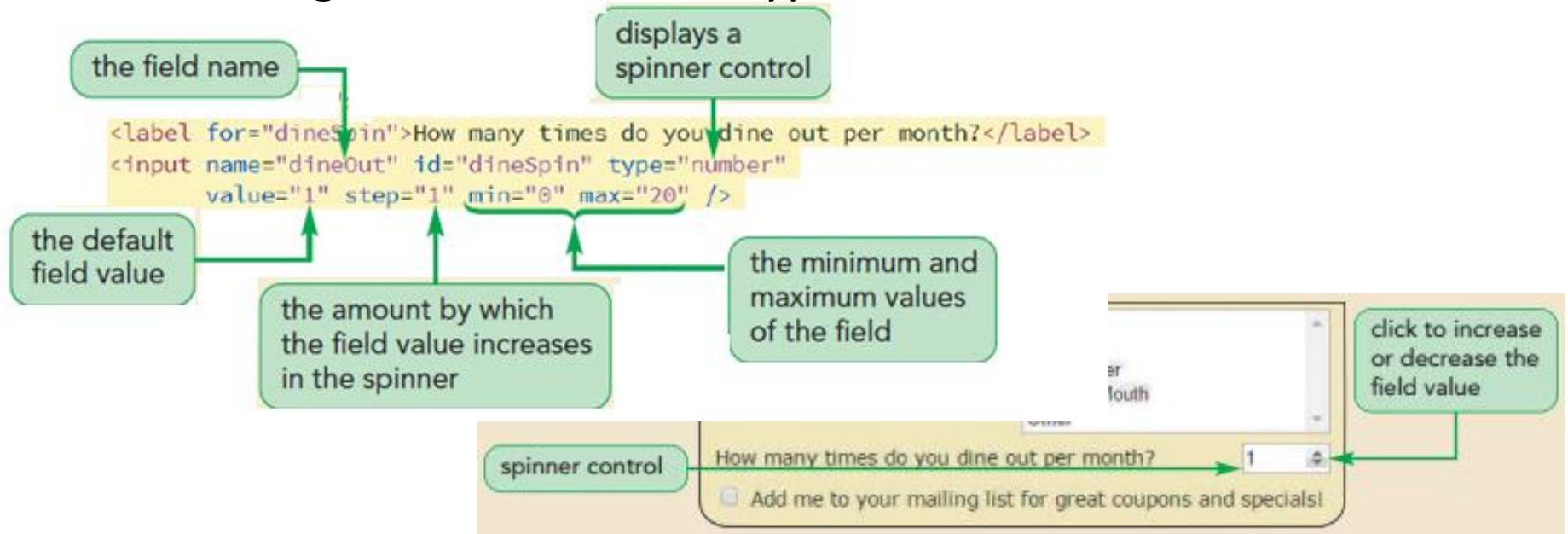
sets the data type of the
visitDate field to "date"

04:21 PM

type="time"

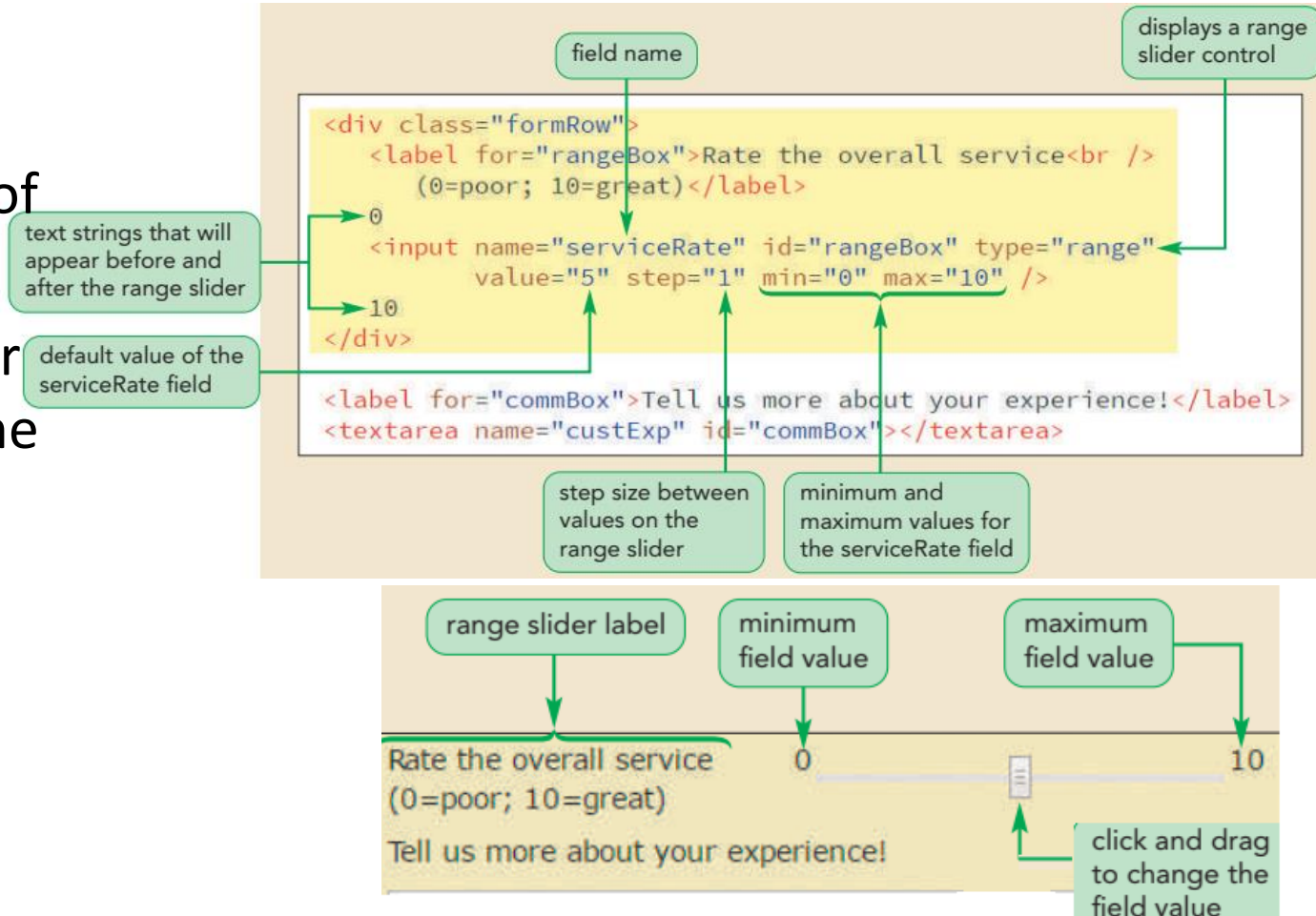
Spinner control - entering numeric values

- **spinner control:** Displays an up or down arrow to increase or decrease the field value by a set amount
 - To create a spinner control, **apply the input element** using the **number** data type



Creating a range slider control

- A **slider control** limits a numeric field to a range of possible values
- To create a slider control, apply the range data type in the input element



The Output element

- A **container element** into which a site or app can inject the results of a calculation or the outcome of a user action
- The `<output>` tag is a newly added tag and was introduced in HTML5

▪ Example 1:

Sum up the value of slider and spin controls



```
<!DOCTYPE html>
<html>
<body>
<p>Sum up the value of slider and spin controls</p>
<form oninput="result.value=parseInt(a.value)+parseInt(b.value)">
  <input type="range" id="b" name="b" value="50" /> +
  <input type="number" id="a" name="a" value="10" /> =
  <output name="result" for="a b">60</output>
</form>
</body>
</html>
```

The progress and meter element



- Progress element
 - Displays an indicator showing the completion progress of a task, typically displayed as a progress bar.
 - It is mostly used to show the progress of activity like a file uploading or downloading on the web.
- Meter element
 - Used to measure data within a given range.
 - It represents either a scalar value within a known range or a fractional value
 - It is also known as a gauge
 - It should be used to display disk usage, voting population etc.

The progress and meter element example



```
<!DOCTYPE html>
<html>
<head>
<title>Output Tag</title>
</head>
<body>
```

Downloading:

```
<progress value="43" max="100">43%</progress>
```

```
<br><br>
```

```
<label for="fuel">Fuel level:</label>
```

```
<meter id="fuel"
  min="0" max="100"
  low="33" high="66"
  optimum="80"
  value="50">
```

at 50/100

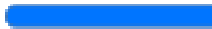
```
</meter>
```

```
</body>
```

```
</html>
```

Value - defines that how much work the task has been completed
Max - defines that how much work the task requires in total

Value - a mandatory attribute which is used to specify a value in numbers (integer or floating point)..
High and low - specifies a range that is considered as high and low value respectively.
Max and min - specifies the maximum and minimum value defined in the range.
Optimum - specifies the optimum value for the gauge

Downloading: 

Fuel level: 

Form Buttons



- Form buttons: A type of form control that performs an action
- Actions performed
 - Run a command from a program linked to the web form
 - Submit the form to a program running on the web server
 - Reset the form fields to their default values
- Types of form buttons
 - Command button
 - Submit button
 - Reset button
 - Custom button

Submit and Reset Button



- Submit button - submits a form to the server for processing when clicked
- Reset button - resets a form, changing all fields to their default values and deleting any field values that a user has entered
- Created using input elements with the type attribute set to “submit” and “reset” respectively
- Example:

```
<input value="text" type="submit" />
```

```
<input value="text" type="reset" />
```
- where text is the string that appears on the button

Command and Custom Buttons



Command button

- Runs a program that affects the content of a page or the actions of a browser
- Created using the input element with the type attribute set to button
- e.g. `<input value="text" onclick="script" type="button" />`
 - text - string that appears on the button
 - script - the name of the program code that is run when the button is clicked

Custom button

- Appearance of a command, submit, and reset button is determined by the browser
- For more control over a button's appearance use the button element
- E.g. `<button type="text"> content </button>`
 - Where type attribute specifies the button type and the content are HTML elements placed within the button

Defining default value and placeholder



```
<label for="city">City</label>
<input name="custCity" id="city" type="text" value="Ormond Beach" />

<label for="state">State</label>
<input name="custState" id="state" type="text" value="FL" />
```

sets the default value
for the custCity field

sets the default value
for the custState field

Default value

- Specified using the value attribute

Placeholder

- A text that appear within a form control, providing a hint about the kind of data that should be entered into a field
- Defined using the placeholder attribute

The diagram shows a form titled "Customer Information" with the following fields and annotations:

- Name***: Placeholder text "first and last name".
- Street address:**: No placeholder.
- City**: Default value "Ormond Beach".
- State**: Default value "FL".
- Postal code**: Placeholder text "nnnnn (-nnnn)".
- Phone number**: Placeholder text "(nnn) nnn-nnnn".
- E-mail***: No placeholder.

Annotations on the left side:

- formatted field set legend (points to the form title)
- default value for the custCity field (points to the City field)
- placeholder text for the phone input box (points to the Phone number field)

Annotations on the right side:

- placeholder text for the name input box (points to the Name field)
- default value for the custState field (points to the State field)
- placeholder text for zip input box (points to the Postal code field)

Form Validation



- Validation: Process of ensuring that a user has supplied valid data
- Types of validation
 - Server-side validation – validation occurs on the web server
 - Client-side validation – validation occurs in the user's browser

What to validate?

- Identifying Required Values
 - The first validation test is to verify if data is supplied for all the required data fields
 - Add the required attribute to the control to identify the required data fields
- Defining the Length of the Field Value
 - For example the syntax to define the maxlength attribute is `<input maxlength="value" />`
 - Example: `<input name="custZip" maxlength="5" />`
 - The maxlength attribute does not distinguish between characters and digits

Form Validation cont'd



- Validating Based on Data Type
 - A form fails the validation test if the data values entered into a field do not match the field type
 - Example:
 - Entering a nonnumeric data for data field with the number type
 - Providing invalid email or url types that does not match the format of a URL
- Testing for a Valid Pattern
 - To test whether a field value follows a valid pattern of characters, test the character string against a regular expression
 - Regular expression or regex is a concise description of a character pattern
 - To validate a text value against a regular expression, add the pattern attribute to the input element.
 - Example: see next slide

Data Validation

(examine the code)

The **pattern** attribute specifies the general pattern that the characters in the field value must follow.

The **min** and **max** attributes define the range of possible field values; the **step** sets the interval between values.

The **datalist** element defines a set of suggested field values.

The **submit** data type creates a button to submit the form for processing.

```
<label for="name">Name *</label>
<input name="custName" id="name" type="text" required />

<label for="phone">Phone number</label>
<input name="custPhone" id="phone" type="tel"
  pattern="^\d{10}$|^\(\(\d{3}\)\)\s*)?\d{3}[\s-]?\d{4}$" />

<label for="dineSpin">How often do you dine out per month? </label>
<input name="dineOut" id="dineSpin" type="number"
  value="1" step="1" min="0" max="20" />

<label for="dish">What's your favorite dish?</label>
<input name="favDish" id="dish" type="text" list="dishType" />
<datalist id="dishType">
  <option value="Big Kahuna Pizza" />
  <option value="BBQ Chicken Pizza" />
  <option value="Pasta Rolls" />
  <option value="Pasto Artichoke Pizza" />
</datalist>

<label for="rangeBox">Rate the overall service at Red Ball<br />
  (0=poor; 10=great)</label>
0 <input name="serviceRate" id="rangeBox" type="range"
  value="5" step="1" min="1" max="10" /> 10

<input type="submit" value="Submit My Survey" />
<input type="reset" value="Cancel" />
```

The **required** attribute indicates that a field value is required.

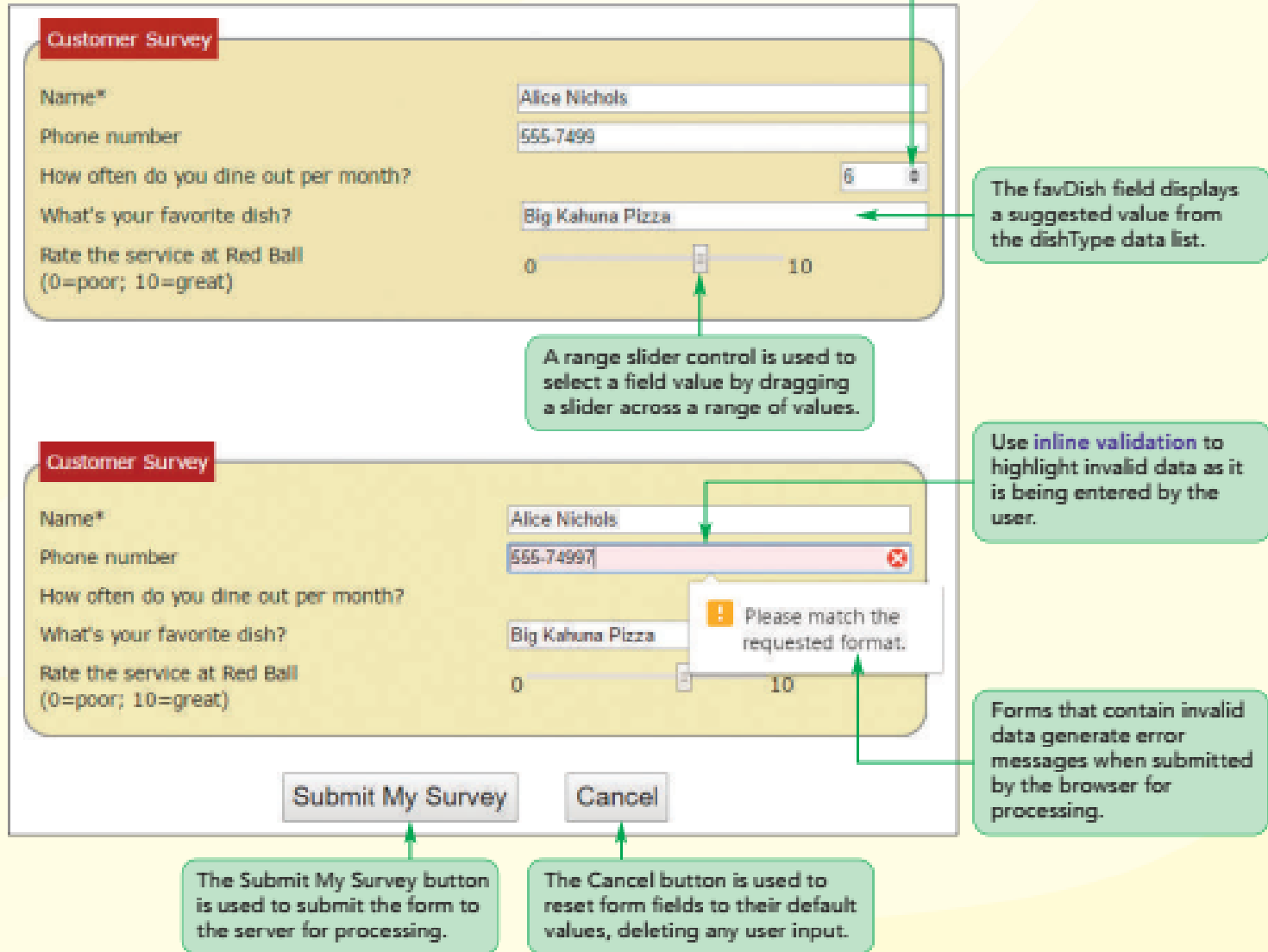
The **number** data type creates a spin box control for data entry.

The **reset** data type creates a button that restores the form to its default values.

The **range** data type creates a range slider for data entry.

Data Validation

(look on the screen)



Customer Survey

Name* Alice Nichols

Phone number 555-7499

How often do you dine out per month? 6

What's your favorite dish? Big Kahuna Pizza

Rate the service at Red Ball (0=poor; 10=great)

A spinner control is used to select a field value by clicking spin arrows to increase or decrease the value by a set amount.

The favDish field displays a suggested value from the dishType data list.

A range slider control is used to select a field value by dragging a slider across a range of values.

Customer Survey

Name* Alice Nichols

Phone number 555-74997

How often do you dine out per month?

What's your favorite dish? Big Kahuna Pizza

Rate the service at Red Ball (0=poor; 10=great)

Please match the requested format.

Use inline validation to highlight invalid data as it is being entered by the user.

Forms that contain invalid data generate error messages when submitted by the browser for processing.

Submit My Survey

Cancel

The Submit My Survey button is used to submit the form to the server for processing.

The Cancel button is used to reset form fields to their default values, deleting any user input.

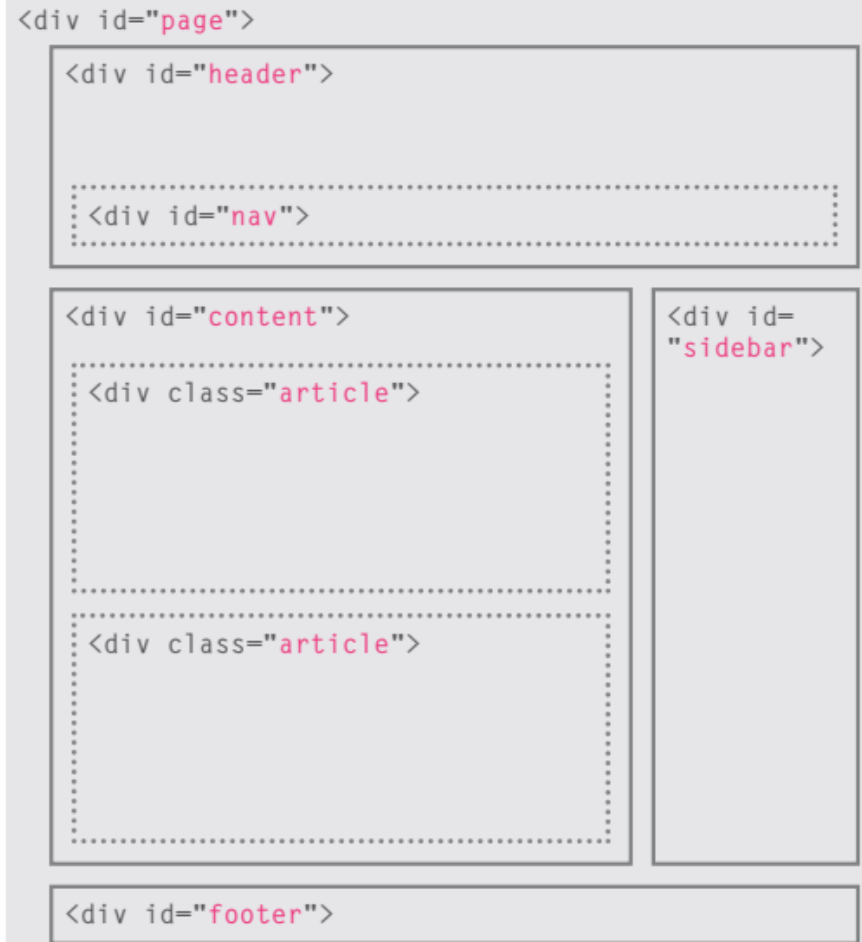
HTML Semantic Elements



<body>

Traditional HTML Layout

- Content sectioning elements allow you to organize the document content into logical pieces.
- For a long time, the `<div>` *element* is used to group together related elements on the web page (such as *a header, an article, footer or sidebar*)
- Web developers are used *class or id attributes* to indicate the *role of the <div> element* in the structure of the page

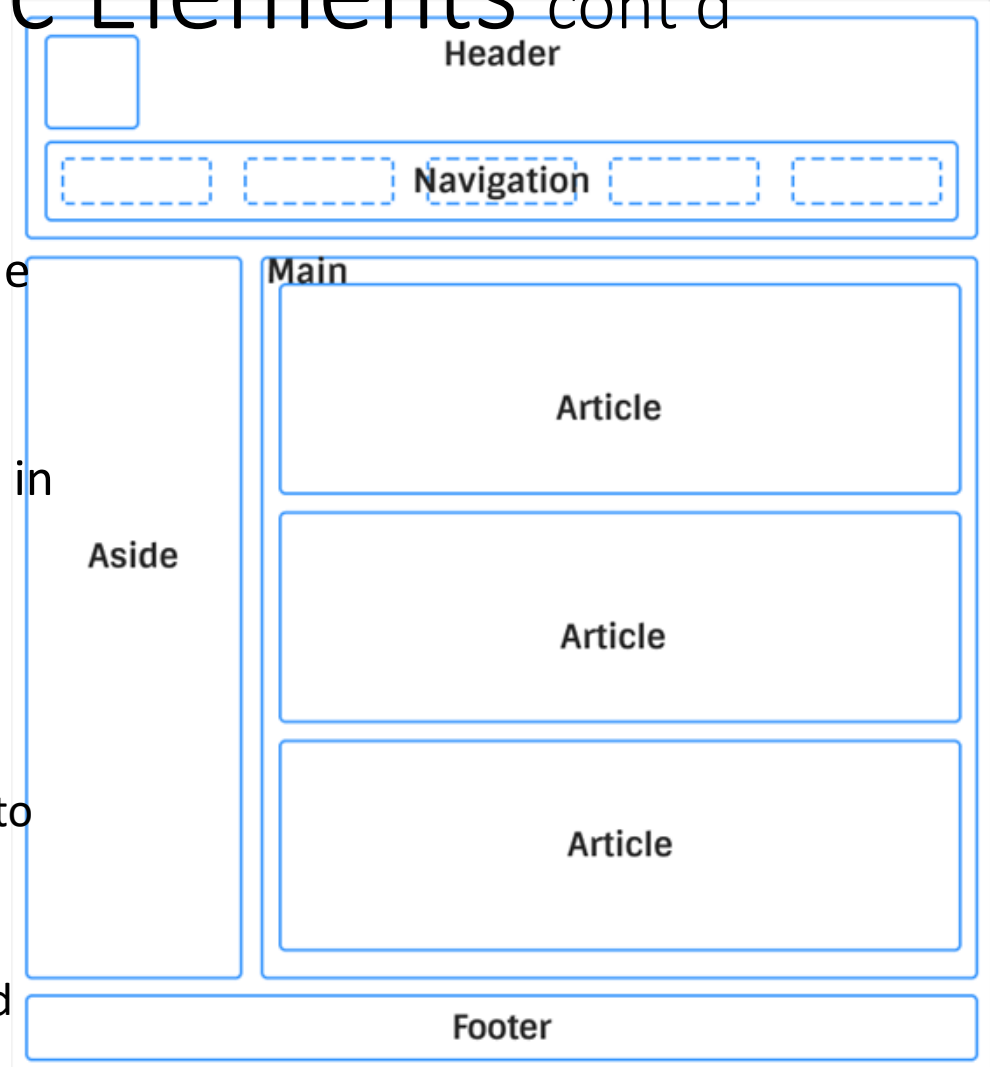


HTML Semantic Elements cont'd



HTML5 Layout Elements

- HTML5 introduces a new set of elements that allow you to divide up the parts of a page
- The names of these elements indicate the kind of content found in them
- HTML5 content sectioning - semantic elements
- **Why semantics?**
 - Clearly describes its meaning to both the browser and the developer
 - Help to structure the code and add readability

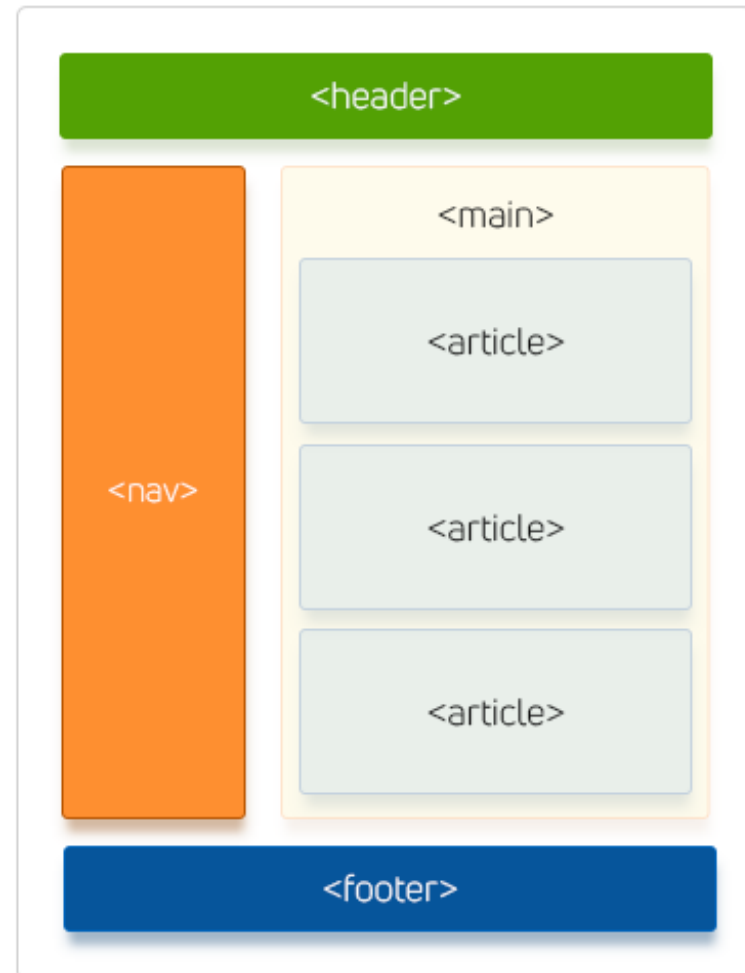
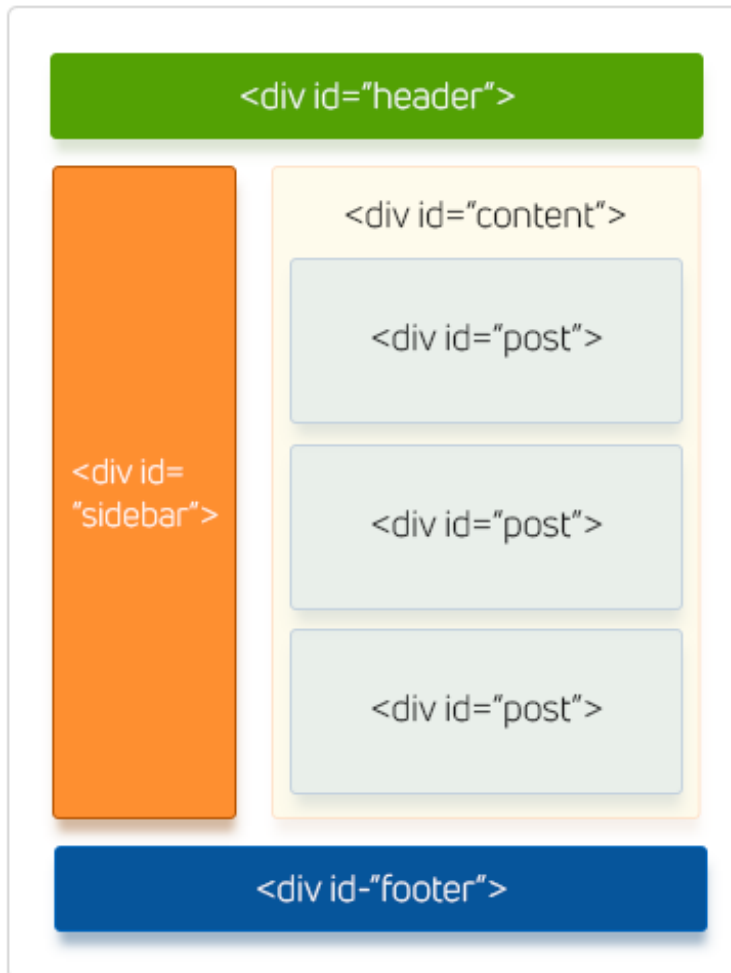


Traditional Vs. HTML5 Page Layout Elements



HTML4: Lost of Classes/id

HTML5: Semantic Tags/Sections



The <header> element



- Specifies a header for a document or a section.
- For introductory content <header> element should be used as container
- Generally it contains one or more heading elements, logo or icons or author's information
- Several <header> elements can be used in one document,
- However, it cannot be placed within a <footer>, <address> or another <header> element.
- It is not sectioning content and therefore does not introduce a new section in the outline
- Intended to usually contain the surrounding section's heading (h1–h6 element)

The <footer> element



- Specifies a footer for a document or a section.
- It is generally used in the last of the section (bottom of the page).
- Typically contains information about the
 - author , contact, and copyright information
 - Sitemap, back to top links,
 - Links to related documents etc.
- Not sectioning content and therefore doesn't introduce a new section.
- To put information like address, e-mail etc. about the author on your web page, all the relevant elements should be included into the footer element.
- For example, enclose information about the author in an <address> element that can be included into the <footer> element.

The <nav> element



- Represents a section of a page whose purpose is to provide navigation links, either within the current document or to other documents
- Common examples of navigation sections are menus, tables of contents, and indexes
- It's not necessary for all links to be contained in a <nav> element
- It is intended only for major block of navigation links; typically the <footer> element often has a list of links.
- A document may have several <nav> elements, for example, one for site navigation and one for intra-page navigation
- User agents, such as screen readers, can use this element to determine whether to omit the initial rendering of navigation-only content

The <main> element



- Represents the dominant content of the <body> of a document.
- It is written within <body> tag
- It is used to accurately describe the primary content of a page.
- Its content should directly related to or expands upon the central topic of a page, or the central functionality of an application.
- Doesn't contribute to the document's outline
- It should be unique to the document and cannot be nested inside other semantic elements like <article>, <aside>, <header>, <footer> elements
- Content that is repeated across other document sections such as sidebars, navigation links, site logos, and search forms shouldn't be included
- If the search form is the main function of the page, it can be included

The <article> element



- Defines an independent self-contained content in a document, page, application or a site
- The content makes sense on its own and intended to be independently distributable or reusable (e.g., in syndication)
- It generally used on Forum post, a magazine or newspaper article, Blog post, News story, a product card, a user-submitted comment, an interactive widget or gadget, or any other independent item of content
- Each <article> should be identified, typically by including a heading (<h1>-<h6> element) as a child of the <article> element
- When an <article> element is nested, the inner element represents an article related to the outer element

The <section> element



- Represents a generic standalone section of a document, which doesn't have a more specific semantic element to represent it.
- Sections should always have a heading, with very few exceptions.
- When you put your content on a web page, it may contain many chapters, headers, footers, or other sections on a web page that is why HTML <section> tag is used.
- Example: A home page could normally be divided into sections for introduction, content, and contact information
- To reiterate, each <section> should be identified, typically by including a heading (<h1>-<h6> element) as a child of the <section> element, wherever possible

The <aside> element



- Represents a portion of a document whose content is only indirectly related to the document's main content
- According to W3C definition, the <aside> element represents content that forms the main textual flow of a document
- It frequently presented as sidebars or call-out boxes
- Do not use the <aside> element to tag parenthesized text, as this kind of text is considered part of the main flow

